The Effect of Public Sector Relocations on Regional Development in Germany

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As regional economic disparities within countries grow, governments are increasingly experimenting with public employment reallocation as a place-based policy. In this paper, I estimate the causal effect on local labor markets of a German policy that relocated about 3,000 public sector jobs to lagging regions. Using novel data on 60 agency relocations from 2015–2025, I estimate employment and population effects for receiving and sending municipalities and use the results to discipline a quantitative spatial model that simulates spatial reallocation and welfare effects. I find that relocations increased private employment shares by up to 2.3% (1.3 percentage points), reduced unemployment by up to 11.9% (0.33 percentage points), and raised population by 1.6%, implying a public-to-private jobs multiplier of 1.08 in receiving locations. Sending locations also see an increase in private sector employment that is rationalized in the model by within-private-sector spillovers exceeding public-to-private sector spillovers.

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1 Introduction

As economic disparities within countries widen, governments in Germany, Denmark, France, Canada, South Korea, the United Kingdom, and Sweden have turned to public employment reallocation as a place-based policy to revive lagging labor markets[1]. Yet economic theory remains ambiguous about its effects. In spatial general-equilibrium models[2], relocating public employment into a region can crowd in private employment by raising local demand, especially for nontradable goods and services. But it can also crowd out private employment if rising wages and housing prices push tradable firms out of the local market. Applied in reverse, the same logic predicts that sending locations, typically national or regional capitals, should lose private employment when public jobs are removed. Yet the literature is largely silent on such effects[3].

To identify what happens in both receiving and sending locations, I ask whether relocations of public-sector jobs to lagging areas improve local labor-market and demographic outcomes in receiving locations, what effects they have on the sending locations from which those jobs are withdrawn, and what the resulting economy-wide general-equilibrium consequences are. I address this question by examining an ongoing public employment relocation policy in Germany.

The German Federal Constitution anchors the objective of equivalent living conditions across regions as a state goal (*GG Art. 72(2)* 1949). Politically, this is interpreted as a mandate to reduce and mitigate disparities between urban and rural areas, as well as across regions in Germany (German Federal Ministry of Agriculture, Food and Regional Identity 2025). In Bavaria, the *Heimatstrategie* represents a state policy response to this mandate. Introduced in 2015, the program aimed to transfer 60 agencies and over 3,155 public-sector jobs and apprenticeships, roughly 1% of all state public-sector jobs, to lagging regions by 2025 (Bavarian Ministry of Finance 2021; Bavarian Ministry of Finance 2025). In addition to reallocating public employment, the program affects the use of scarce urban space. The planned relocations are expected to release around 37,000 square meters of office floor space in prime locations of the urban sending areas (Bavarian Ministry of Finance 2026).

From an empirical perspective, the program offers several features that make it well-suited to studying the effects of public-employment relocation. An empirical analysis of public-employment relocation is especially informative when the policy generates sizable spatial shifts in public employment, treatment assignment follows observable rules, and implementation covers a sufficiently broad set of locations. The Bavarian program offers these features because it reallocates public employment across many municipalities, thereby allowing me to study a policy that targets a broad set of places rather than a single relocation episode.[4] Bavaria allocates relocations using a quantitative eligibility index

[1] See Freitas (2025) for an overview on past and current public employment reallocation programs.

[2] See Helpman (1998) for a detailed exposition.

[3] One notable exception is Faggio (2019), who reports little adverse impact on places that lose public jobs in the UK.

[4] Related evidence largely comes from large, concentrated "big push" episodes, including capital-city relocations such as the Berlin–Bonn case (Becker et al. 2021; Faggio, Schluter, et al. 2022) and large

that ranks municipalities by socioeconomic distress. In addition, agencies are mostly assigned only to officially designated mid-sized centers, *Mittelzentren*, in qualifying districts, reflecting spatial-planning requirements. These institutional features create a setting in which the policy target is ex ante explicit, treatment is allocated according to observable criteria, and implementation is distributed across many municipalities.

These advantages notwithstanding, the setting still raises three key empirical challenges. The first is selection bias, the second is treatment-time heterogeneity, and the third is external validity. I address selection bias by reconstructing the government's allocation rule and applying it to all German municipalities to create an "eligible-but-untreated" pool outside Bavaria. I then form a control group using Mahalanobis matching on the rule's components and pre-policy outcomes, and implement difference-in-differences estimators that are valid under selection on observables. The staggered rollout between 2015 and 2025 introduces treatment-time heterogeneity, which I address by combining long differences with the Callaway–Sant'Anna estimator aggregated by event time. Finally, to move beyond the partial-equilibrium focus in the literature, I estimate effects for both receiving and sending locations empirically and rationalize the resulting spatial general-equilibrium adjustments in a two-sector quantitative spatial model.

I find that the program increased private employment, population, and the working-age population in receiving municipalities, while also reducing unemployment. After netting out the mechanical addition of relocated public jobs, I find an increase in private-sector employment shares by 1.1% or 0.65 percentage points following the start of relocations in 2015, rising to as much as 2.3% or 1.3 percentage points by 2021. These findings are statistically significant for the period spanning 2019–2022. These results correspond to a public-to-private job multiplier of 1.08 jobs per public job relocated.

The relocation initiative led to a decrease in unemployment rates in all subsequent years after the program began, except 2016. The empirical findings show statistically significant reductions in unemployment for 2017 and 2018. Specifically, the unemployment rate declined by 5.6% or 0.20 percentage points in 2017 and by 11.9% or 0.33 percentage points in 2018 relative to 2014.

Treated municipalities experienced an increase in local population of 1.2% in 2017, followed by a further rise of 1.6% in 2018. The effect on the working-age population share was even more pronounced, reaching 2.1% in 2018. This upward trajectory was notably disrupted during and after the pandemic period.

Sending locations were not harmed by the policy. Instead, their employment share rose by 0.77 percentage points, unemployment fell by 0.59 percentage points, and the working-age population increased by 1.11 percentage points.

My empirical results are robust to the composition of the control group. I construct a Bavaria-only control by restricting the analysis to municipalities that are never eligible or never treated within Bavaria and reweighting the estimations based on pre-policy outcomes. The resulting estimates closely track the baseline, indicating that findings are not an artifact of idiosyncratic developments in Bavaria. I re-estimate effects using

single-agency moves from capital cities to inland regions, such as the BBC's partial relocation from London to Salford (Nathan et al. 2024).

interaction-weighted two-way fixed effects (Sun and Abraham 2021), obtaining magnitudes and dynamics consistent with the baseline Callaway–Sant'Anna estimates. Event-study graphs show flat pre-trends.

While these results are robust, the reduced-form evidence leaves several important questions open. It does not show how municipalities that neither send nor receive public employment adjust. It cannot speak to the implications for aggregate output, welfare, or whether any groups of municipalities are net losers from the policy. As public employees leave, the resulting decline in local population should reduce demand for non-traded goods and services, such as hairdressers, coffee shops, and local retail, rather than increase it as the empirical results suggest. If a contraction in public jobs in thriving areas such as the state capital Munich creates space for private-sector growth, the empirical results may be consistent with a richer spatial equilibrium.

To formalize this mechanism, I build a quantitative spatial model of Germany with a private sector and a state public sector. The model captures a simple misallocation logic. Public-sector jobs in this setting provide traded administrative services, so their baseline productivity depends less on local conditions than private-sector productivity does. Since private-sector productivity is highest in urban sending locations such as Munich and lower in receiving locations, reallocating a fixed number of public jobs from the former to the latter can reduce the opportunity cost of public employment and improve the spatial allocation of economic activity. The model also allows private-sector productivity to depend on sectoral spillovers, so that receiving locations benefit from the local demand shock, while sending locations can gain as labor and, potentially, scarce urban space are reallocated toward private activity with stronger agglomeration externalities. I implement the public-employment reallocation policy as a counterfactual scenario and compare the resulting spatial equilibrium to the empirical estimates. The model rationalizes why both receiving and sending locations can experience employment gains.[5]

My paper contributes to three strands of literature. First, it contributes to the literature on place-based policies, which studies whether spatially targeted interventions can improve outcomes in lagging regions (Moretti 2012; Kline and Moretti 2014; Ehrlich and Seidel 2018; Duranton and Venables 2018; Ehrlich and Overman 2020). I study public employment reallocation as a distinct place-based policy instrument. Unlike subsidies, tax incentives, or infrastructure programs, this policy directly changes the spatial allocation of public-sector employment. The Bavarian setting also allows me to address a central empirical challenge in this literature. Since treatment assignment follows an explicit policy allocation rule, I can use the rule's components to construct a credible control group and reduce concerns about selection into treatment.

Second, the paper contributes to the literature on public employment and local labor markets. Existing work asks whether public jobs crowd in or crowd out private employment, with evidence from public-sector expansions, contractions, and relocations (Faggio and Overman 2014; Senftleben-König 2014; Auricchio et al. 2020; Faggio 2019; Jofre-Monseny et al. 2020; Lee et al. 2024; Becker et al. 2021; Faggio, Schluter, et al.

[5] As a next step, I will estimate the model's agglomeration spillover elasticities via GMM, targeting the empirical policy responses as moment conditions.

2022). A further contribution is the construction of what is, to my knowledge, the first yearly municipal employment panel for Germany that combines the universe of social-security records from the IAB Integrated Employment Biographies with the universe of civil servants by workplace and place of residence.[6] This panel will measure total local employment and provide a finer sectoral breakdown than the German Regional Statistical Database. I show that public employment relocation increases private employment in receiving locations without harming sending locations.

Third, the paper contributes to the literature that incorporates public employment into spatial equilibrium models (**guillouzouic2021**; Becker et al. 2021; Guillouzouic et al. 2024). I build a two-sector quantitative spatial model with a private sector and a state public sector. The model rationalizes employment gains in both origins and destinations through a combination of spatial misallocation and asymmetric sectoral spillovers. Public jobs in this setting provide traded administrative services, while private-sector productivity varies strongly across space. Reallocating public employment from high-productivity urban origins to lower-productivity receiving locations can therefore reduce the opportunity cost of public employment. At the same time, stronger within-private-sector spillovers imply that sending locations can gain when labor and scarce urban space are reallocated toward private activity.

The paper proceeds as follows. I first describe the institutional setting and the policy program. I then present the data, including the novel relocation dataset and administrative sources. Next, I outline the matching procedure and empirical methodology, then present reduced-form results for labor markets, population, and sending locations. I then introduce a two-sector quantitative spatial model, describe its setup and calibration, and present a counterfactual simulation of the public employment reallocation policy. The final section concludes.

2 Institutional Setting

Germany has relocated public jobs for decades. After World War II, it moved the capital from Berlin to Bonn. In 1999, it moved the seat of government back to Berlin and shifted several ministries to the new eastern states (German Bundestag 1992).[7] Germany currently runs two major relocation policies.[8]

Federal Program (Energy Transition). To compensate coal-producing regions for the energy transition, the federal government enacted the *Strukturstärkungsgesetz Kohleregionen*. The program aims to relocate about 5,000 federal jobs by 2028 and to deploy a €40 million subsidy fund for affected areas (Press and Information Office of the Federal Government 2023). The rollout is still pending, and most funds remain

[6] I thank Wolfgang Dauth, Nina Gläser, and the IAB REGIO group for their support in accessing the IAB data.

[7] See Becker et al. (2021) on the move from Berlin to Bonn and Faggio, Schluter, et al. (2022) on the return to Berlin.

[8] To document the scope of current public employment relocation policies in Germany, I conducted a structured search on Nexis Uni (LexisNexis), identifying press reports on 22 federal and 87 state agencies from 2005 to 2032. Appendix A details sources and coding.

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Bavarian Program (Heimatstrategie). In 2014, Bavaria launched the *Heimatstrategie*. From 2015 to 2025, the state planned to decentralize 60 agencies and to transfer more than 3,155 public-sector jobs and apprenticeships, corresponding to roughly 1% of state public-sector employment, to lagging regions. The policy combines relocations from existing sending locations, the creation of new regional branches, and expansions of existing agencies. Most relocated jobs originate in Munich, while smaller numbers are moved from Nuremberg and other urban sending locations. In addition to reallocating public employment, the program affects the use of scarce urban space. The planned relocations are expected to release around 37,000 square meters of office floor space in prime locations of the urban sending areas (Bavarian Ministry of Finance 2026).

The main objective of the policy is to establish equivalent living conditions between urban areas such as the state capital Munich and lagging regions, mainly in northern Bavaria and near the border regions with the Czech Republic and Austria. Despite being one of the most economically successful states in Germany, Bavaria exhibits substantial intraregional disparities. It contains municipalities that rank among both the highest and lowest in the German income distribution. In Gruenwald, located immediately south of Munich, mean income per taxpayer exceeded €136,000 in 2014. By contrast, in Simbach, located at the Austrian border, mean income per taxpayer was €19,637 in the same year (German Regional Database 2024).

In December 2021, the Bavarian government extended the program through *Heimatstrategie Stufe 2*, which runs until 2032. The extension reallocates fewer agencies but more jobs, with 15 agencies and 2,720 jobs slated for redistribution to 13 municipalities in 11 districts.

In addition to these measures, supplementary relocations were initiated in 2016 to support Middle Franconia and Lower Bavaria specifically. Because these relocations were targeted to particular regions rather than selected through the structural index that identifies areas with particular need for action, I exclude them from the analysis. Table 1 summarizes all relocation programs running in Bavaria and the corresponding allocation rules. My analysis focuses on *Heimatstrategie Stufe 1*.

The site-selection criteria combine quantitative and qualitative elements. A structural index flags distressed districts using administrative data on five indicators: (i) population projections, (ii) unemployment, (iii) employment density, (iv) income per taxpayer, and (v) net migration of 18–30-year-olds. Table 2 reports the indicators, their administrative data sources, and their weights in the index. The Bavarian government normalizes the weighted index to 100 at the Bavarian mean and sets 90 as the eligibility cutoff. Districts with an index value of 90 or below are eligible for the public employment reallocation program. Municipalities below the cutoff qualify even if their district lies above it.[9]

Beyond the index, receiving sites must be designated as *Mittelzentren* in the Regional Development Plan and must not already receive major place-based funds, for example

[9] About 88% of relocation projects are directed toward areas identified through the structural index. The remainder includes six conversion municipalities affected by defense-base closures in 2011, when Germany suspended compulsory military service.

Phase	Program Name	Allocation Rule
I	*Heimatstrategie.* Regionalization of the Administration	Distressed areas as of the Regional Development Plan 2013/14*
II	*Heimatstrategie.* Administration Relocation Bavaria 2030	Distressed areas as of the Regional Development Plan 2018**
Other	Structural Concept "Opportunities in the Whole State"	Regionally bound to Middle Franconia and Lower Bavaria
Other	Additional single regionalization measures	No explicit allocation rule

Table 1. Policy background on public-sector relocations in Bavaria. In this analysis, I only include relocations from the first phase of the *Heimatstrategie.*
*Based on the Bavarian Regional Development Plans 2014 and 2018.

through concurrent university initiatives. Figure 1 maps treatment status.

The state government selected agencies for relocation based on the spatial scope of their activities. The state selected agencies with supra-regional mandates under the jurisdiction of the Free State of Bavaria and excluded inherently local service providers such as police stations and schools. Before relocation, these agencies were typically located in densely populated areas.

Relocation costs are usually opaque, but Bavaria published an itemized cost report after a parliamentary inquiry (Kaltenhauser 2023). Table 3 reports the different cost categories for the relocation program up to 2022. Personnel costs from 2016 to 2022 amount to €16.6 million. They mainly consist of incentive payments that the Bavarian state government offers employees willing to relocate. These include mobility premia, a one-time lump-sum payment of €3,000 for permanent relocation, reimbursement of additional moving costs, expense reimbursement for employees commuting between locations or staying near the new workplace during the week, and separation allowances to cover supplementary costs when an employee is temporarily separated from the main residence due to relocation.

Construction expenditures total €74.9 million. Renovation expenses across locations amount to €3.7 million. Moving costs from 2015 to 2021 total €0.45 million. Rental costs from 2015 to 2022 total €18.2 million. The cumulative financial outlay across all categories is roughly €113.9 million (ibid.).

A back-of-the-envelope division yields roughly €38,000 per job or apprenticeship relocated so far. Because fixed costs are front-loaded and support future placements, the average cost per relocated position should fall as additional moves occur.

Weight	Indicator	Statistic	Source
30%	Population projections	Change between 2009 and 2021	GRD, BSO
30%	Unemployed persons	2012	GRD
10%	Employment density per 1,000 inhabitants	2012	GRD, FEA
20%	Income per taxpayer	2007	BSO
10%	Net migration of 18–30-year-olds per 1,000 inhabitants	2011	GRD, BSO

Table 2. Weights and composition of the structural index at the municipal level. GRD denotes the German Regional Database, FEA denotes the Federal Employment Agency, and BSO denotes the Bavarian Statistical Office.

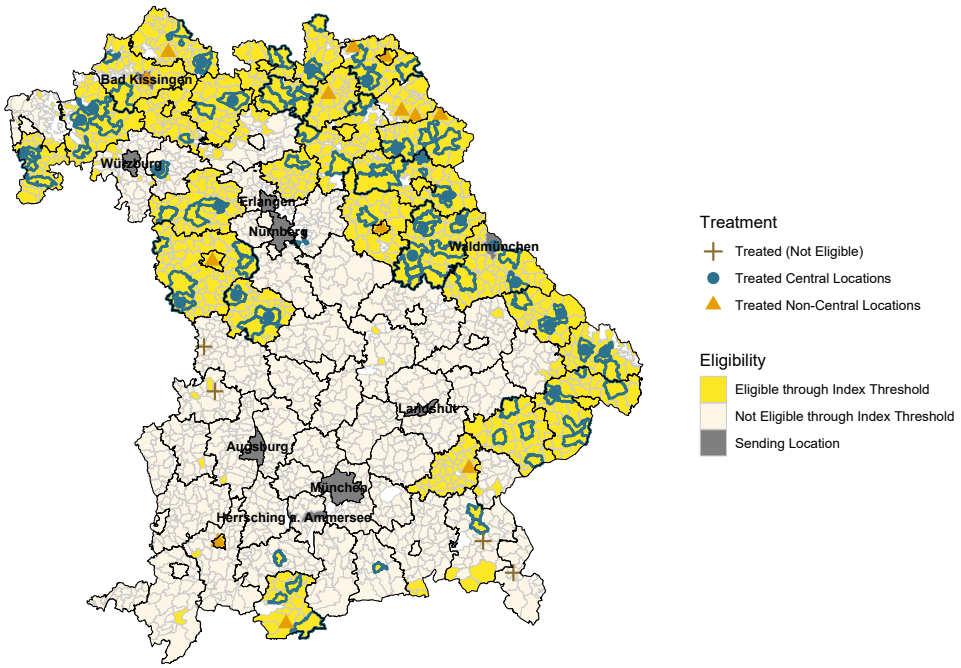


Figure 1. Spatial scope of the Bavarian relocation program. Municipalities eligible through the quantitative structural index, defined as districts or municipalities at 90% or less of the Bavarian average, are shown in yellow. Non-eligible municipalities are shown in off-white. Mid-sized central locations are indicated by a teal outline. Treatment is shown by symbols: treated central locations in teal, treated non-central locations in orange, and treated but not eligible locations in brown. Sending locations are marked in gray. Treated but ineligible municipalities are locations that lost defense employment in 2011 when Germany suspended compulsory military service.

Cost Category	Total Cost (€)
Personnel costs (2016–2022)	16,623,849.00
Construction costs	74,923,320.64
Renovation costs	3,708,003.37
Moving costs (2015–2021)	450,713.00
Rental costs (2015–2022)	18,188,015.00
Total	113,893,901.01

Note: This table summarizes costs across multiple years: personnel costs from 2016 to 2022, construction costs for four sites, moving costs from 2015 to 2021, and rental costs from 2015 to 2022.

Table 3. Preliminary costs of the Bavarian public employment reallocation program, 2015–2022.

3 Data

3.1 Data on the Relocation Program

I assemble data on Bavaria's relocation program from the BayernAtlas open-data portal (State Office for Digitization, Broadband and Geodetic Measurement 2024). For each relocation case, I record the program name, lead ministry, agency address, number of jobs relocated, and relocation status. I complete missing completion dates from the annual *Heimatstrategie* reports and, when needed, from searches in local newspapers. On average, a relocation corresponds to 1.3% of total employment in the receiving municipality. Fig. 21 in Appendix B shows that origin–destination distances cluster between 150 and 250 km, implying a mean road travel time of about 126 minutes. The full dataset is available on request.

3.2 Employment Data

A central data contribution of this paper is the construction of a yearly municipal employment panel for Germany that measures total employment by workplace and place of residence. The final panel will combine the universe of social-security records from the IAB Integrated Employment Biographies with the universe of civil servants in Germany at the municipal level.[10] This will allow me to measure total local employment rather than only employment covered by social security, and to use a finer sectoral breakdown than is available in the German Regional Statistical Database. This data construction is currently in progress.

For the empirical results in the current version of the paper, I use annual municipal employment for 2008–2024 from the German Regional Database (GRD), compiled from Federal Employment Agency (FEA) records (German Regional Database 2024). These data include employment headcounts by place of residence, by workplace, and for residents who also work in their home municipality. They cover employees subject to social-security

[10] I thank Wolfgang Dauth, Nina Gläser, and the IAB REGIO group for their support in accessing the IAB data.

contributions, including private-sector employees and public-sector employees covered by social security, but they do not include civil servants. Table 6 reports descriptive statistics for the outcomes by control group used in the empirical analysis.

Since the current empirical data do not yet separately identify private and public employment at the yearly municipal level, I take a conservative approach to the mechanical effect of the policy. In the long-difference and event-study specifications, I deduct the number of relocated public jobs from employment outcomes in treated municipalities. The resulting estimates should therefore be interpreted as net employment responses after removing the direct accounting effect of the relocated jobs themselves. This adjustment is preliminary and will be replaced once the combined IAB and civil-servant panel is complete.

For the quantitative spatial model, I use the 2011 German Census as a preliminary solution. The Census is useful for the model because it reports employment by municipality, sector, workplace, and place of residence. It is also the closest pre-treatment source with this sectoral structure, since the reference date is May 9, 2011, four years before the introduction of the *Heimatstrategie*. The limitation is that the Census does not provide a yearly panel, so it cannot be used directly for the reduced-form event-study analysis.

To construct employment in the public and private sectors for the QSM model calibration, I rely on commuting data from the 2011 German Census. I use the in-commuting tables by sector, code `2000X_2041`, and the out-commuting tables by sector, code `2000X_2045`, from the Census database. Both tables are available at the municipal level as a sample.

Workplace employment is constructed from the in-commuting table by filtering for the category *"Einpendelnde in Arbeitsortgemeinde"*, meaning in-commuters to the workplace municipality. Residence employment is constructed from the out-commuting table by filtering for the category *"Arbeitsort in Hauptwohnsitzgemeinde"*, meaning place of work in the main municipality of residence. This category includes workers who are *"Binnenpendler innerhalb der Hauptwohnsitzgemeinde"*, meaning commuters within their main municipality of residence, workers who are *"überwiegend zuhause"*, meaning working predominantly at home, and *"Auspendler aus der Hauptwohnsitzgemeinde"*, meaning out-commuters from the main place of residence.

To define public-sector employment in the Census, I use the sectoral category *"öffentliche Verwaltung, Verteidigung, Sozialversicherung"*, meaning public administration, defense, and social security. All other sectors are classified as private employment. Applying this taxonomy gives the spatial distribution of private employment shown in **??**. **??** depicts the distribution of public employment in Germany.

The German sample and the Bavarian subsample show similar average magnitudes of workplace employment, with mean values of roughly 59,000 jobs in both cases, as shown in Table 4 and Table 5. The composition of workplace employment is also stable across the two samples. Public employment accounts for around 10 percent, and private employment for around 90 percent. The more pronounced differences appear on the residence side. Average residence employment is substantially lower in Bavaria, at 25,600, than in the German sample, at 34,000, reflecting the greater role of commuting into Bavarian municipalities. Correspondingly, the maximum number of residents employed

Variable	Mean	SD	Min	P25	P75	Max
Total Workplace Employment (1,000s)	58.48	149.66	1.12	14.22	49.80	2135.30
Public Workplace Employment (1,000s)	6.38	16.03	0.00	1.22	5.62	228.94
Private Workplace Employment (1,000s)	52.10	134.08	1.12	12.79	44.50	1919.60
Total Residence Employment (1,000s)	33.99	158.19	1.65	6.37	20.59	4297.12
Public Residence Employment (1,000s)	4.17	20.78	0.17	0.61	2.40	612.53
Private Residence Employment (1,000s)	29.82	137.54	1.39	5.77	18.16	3684.59
Public Workplace Employment Share	10.08	4.80	0.00	7.04	13.29	28.37
Private Workplace Employment Share	89.92	4.80	71.63	86.71	92.96	100.00
Public Residence Employment Share	10.87	2.60	4.26	9.00	12.64	21.48
Private Residence Employment Share	89.13	2.60	78.52	87.36	91.00	95.74

Table 4. Summary statistics of workplace and residence employment variables for the sample of German municipalities (N = 1600).

in a single municipality is much smaller in Bavaria, at 1.8 million, than in the German sample, at 4.3 million. The shares of public and private residence employment are comparable, with public employment representing about 11 percent in the German sample and 10.5 percent in Bavaria. Variability, as measured by the standard deviation, is consistently lower in the Bavarian subsample, which reflects both the smaller number of municipalities and the reduced heterogeneity relative to the national distribution.

3.3 Unemployment Data

I use annual municipal unemployment for 2008–2024 from the German Regional Database (GRD), which compiles Federal Employment Agency (FEA) records, to estimate long-difference and event-study models (German Regional Database 2024).

3.4 Controls Used in Mahalanobis Matching

I draw the matching controls from the German Regional Database. These controls include total population, working-age population, disposable income, and net migration. Population outcomes are observed for the years 2008 to 2023. More current data are not available yet.

3.5 Travel Costs

I obtain commuting flows at the municipal level from the Bundesagentur für Arbeit. Pairwise travel times between Bavarian municipalities are computed in R using the `osrm` package, which interfaces with OpenStreetMap routing data. The procedure requests car-routing durations along the road network for every ordered municipality pair (i, j) and returns a matrix of shortest-path times in minutes, denoted τ_{ij}. The routing profile follows the package defaults for private vehicles, so τ_{ij} captures network connectivity and

Variable	Mean	SD	Min	P25	P75	Max
Total Workplace Employment (1,000s)	59.81	151.93	2.54	18.17	51.47	2135.30
Public Workplace Employment (1,000s)	6.24	15.82	0.00	1.53	5.48	215.70
Private Workplace Employment (1,000s)	53.57	136.35	2.54	16.59	44.65	1919.60
Total Residence Employment (1,000s)	25.59	117.23	2.16	6.30	16.14	1816.58
Public Residence Employment (1,000s)	2.96	13.86	0.20	0.63	1.76	213.91
Private Residence Employment (1,000s)	22.63	103.39	1.90	5.67	14.46	1602.67
Public Workplace Employment Share	10.07	4.46	0.00	7.00	13.02	26.69
Private Workplace Employment Share	89.93	4.46	73.31	86.98	93.00	100.00
Public Residence Employment Share	10.53	2.55	5.56	8.64	12.14	21.48
Private Residence Employment Share	89.47	2.55	78.52	87.86	91.35	94.44

Table 5. Summary statistics of workplace and residence employment variables for the sample of Bavarian municipalities (N = 270).

congestion-related penalties implicit in the underlying OSRM engine. Diagonal elements are set to zero, $\tau_{ii} = 0$. Although τ_{ij} need not equal τ_{ji} because one-way streets and network asymmetries can induce directional differences, only the former enters the travel costs in the model.

Municipality-level summaries used for descriptive maps are obtained by averaging τ_{ij} across all destinations j for each origin i. The resulting mean travel times provide a compact measure of network centrality within Bavaria and are visualized in **??**. The full $\{\tau_{ij}\}$ matrix enters the model via commuting costs $d_{ij} = \exp(\kappa\, \tau_{ij})$.

3.6 Rents, Housing Prices, and Commercial Floor Space

I use a rent price index as a proxy for floor-space prices available via the GitHub toolkit[11] for Ahlfeldt, Heblich, et al. (2023). The R code is available at `https://github.com/dndfreitas/ps_emp_qsm` and can be accessed upon request.

I will extend the data section by adding commercial floor-space data based on the micro-geographic property price and rent indices developed by **ahlfeldt2026**. This extension will allow the model to incorporate the real-estate margin of the policy, in particular the release of office floor space in prime urban sending locations.

[11] `https://github.com/Ahlfeldt/AHS2023-toolkit`

	BY Control			DE Control			Treated		
Variable	N	Mean	SD	N	Mean	SD	N	Mean	SD
Outcomes									
Log Residence Employment	38	7.2	0.98	38	7.4	1.5	38	8.4	0.67
Residence Employment Share	38	60	3.5	38	59	4.2	38	58	3.4
Log Workplace Employment	38	6.6	1.5	38	6.4	2.8	38	8.6	0.83
Workplace Employment Share	38	39	28	38	38	26	38	77	25
Log Employment (Res. = Workp.)	38	5.2	1.4	35	5.8	2.1	38	7.6	0.89
Employment (Res. = Workp.) Share	38	9.4	5.8	35	13	8.9	38	26	7.5
Log Unemployment	38	4	1.1	38	4.5	1.7	38	5.6	0.81
Unemployment Share	38	2.5	0.78	38	3.4	1.5	38	3.7	0.98
Log Total Population	38	8.2	0.99	38	8.4	1.5	38	9.4	0.66
Total Population (1000s)	38	6	8.1	38	10	12	38	15	11
Log Working-Age Pop.	38	7.8	0.98	38	7.9	1.5	38	9	0.66
Working-Age Pop. Share	38	67	2.1	38	66	2	38	64	2.6

Table 6. Descriptive statistics: outcomes by control group. All outcome variables correspond to the year 2014. DE Control corresponds to the control group of municipalities outside the State of Bavaria. BY Control corresponds to the control group of municipalities inside the State of Bavaria. Res. = residence, Workp. = workplace, Pop. = population, Unemp. = unemployment, Inc. = income.

4 Empirical Methodology

4.1 Long Differences

I combine difference-in-differences estimation with Mahalanobis distance matching. The estimating equation for the long-difference analysis is

$$\Delta Y_{ik} = \delta_p + \beta T_i + \gamma X_i' + \epsilon_i,$$

where ΔY_{ik} denotes the log difference in local labor-market outcome k, such as the employment share, working-age population share, or unemployment share, for municipality i between 2022 and 2014.[12] The term δ_p denotes pair fixed effects for each nearest-neighbor match based on Mahalanobis distance. T_i is the treatment indicator, and X_i' is a vector of municipality characteristics based on the structural index used by policymakers to determine treatment eligibility.

Causal interpretation requires parallel trends and no anticipation. The parallel trends assumption requires that, absent the public-sector relocation, treated Bavarian municipalities would have followed the same trajectory as matched control municipalities in local labor-market and population outcomes. The no-anticipation assumption requires that outcomes were not affected by the impending treatment in pre-treatment periods. The announcement in 2015 and the staggered implementation of relocations from 2015 to 2025 could introduce anticipation effects. To assess this possibility, I estimate event-study specifications, which do not reveal significant pre-treatment effects. This absence of

[12]For population outcomes, I use 2023 as the endpoint.

anticipation effects is consistent with private-sector firms waiting until local demand changes materialize and with public employees relocating only when new agencies open.

The Mahalanobis distance is defined as

$$D^2 = (X - Y)' S^{-1} (X - Y),$$

where X and Y represent the vectors of observable covariates for treated and control municipalities, respectively. The inverse covariance matrix S^{-1} weights the distance calculation by the covariance structure of the matching variables. This gives relatively more weight to differences along dimensions with lower variance and accounts for correlations between covariates.

I identify the nearest control unit for each treated municipality. Exact matching is infeasible because matching on several continuous covariates quickly makes the comparison space sparse.[13] I match 38 treated municipalities to 38 control counterparts from outside Bavaria. I use the `MatchIt` package in R to implement Mahalanobis nearest-neighbor matching, which approximates a blocked comparison design.[14]

I examine municipalities within and outside Bavaria to select an appropriate control group.

Boxplots in Fig. 2 show that both municipalities inside Bavaria and municipalities outside Bavaria form plausible control pools based on the distribution of matching covariates. Yet balance plots in Fig. 3 and Fig. 22 in Appendix C reveal superior matching quality with municipalities outside Bavaria, particularly in terms of the variance ratio between treated and control groups, which is closer to one.

Importantly, selecting controls from outside Bavaria mitigates potential spatial spillover effects, a risk underscored by the proximity of treated municipalities within Bavaria, as shown in Fig. 4. It also reduces the risk of including a municipality that received a relocation through one of the other Bavarian relocation programs described in Table 1.

Post-matching improvements are particularly notable for employment and the young population, as shown in Tables 7 and 8. This indicates that the matching procedure substantially improves the alignment of covariate distributions between treated and control municipalities.

4.2 Staggered Difference-in-Differences

I implement event-study estimators to explore the time-varying effects of the policy. The staggered timing of the policy rollout introduces treatment-time heterogeneity, which has been widely discussed in the difference-in-differences literature. A common concern is that early-treated units can receive negative weights when they serve as controls for later-treated units. To address this issue, I use the estimators proposed by Callaway

[13] This is the usual curse of dimensionality in matching problems. As the number of covariates increases, it becomes increasingly difficult to find untreated units that are close to treated units along all dimensions.

[14] In a blocked experiment, treatment and control units are compared within groups of units with similar observable characteristics. Mahalanobis matching approximates this logic in an observational setting by pairing treated and control municipalities with similar pre-treatment covariates.

Figure 2. Boxplots showing the distribution of covariates used for matching across treated and possible control groups. Teal: treated municipalities. Dark blue: control-group municipalities in Bavaria. Orange: control-group municipalities outside Bavaria. Grey: other Bavarian municipalities. White: other German municipalities.

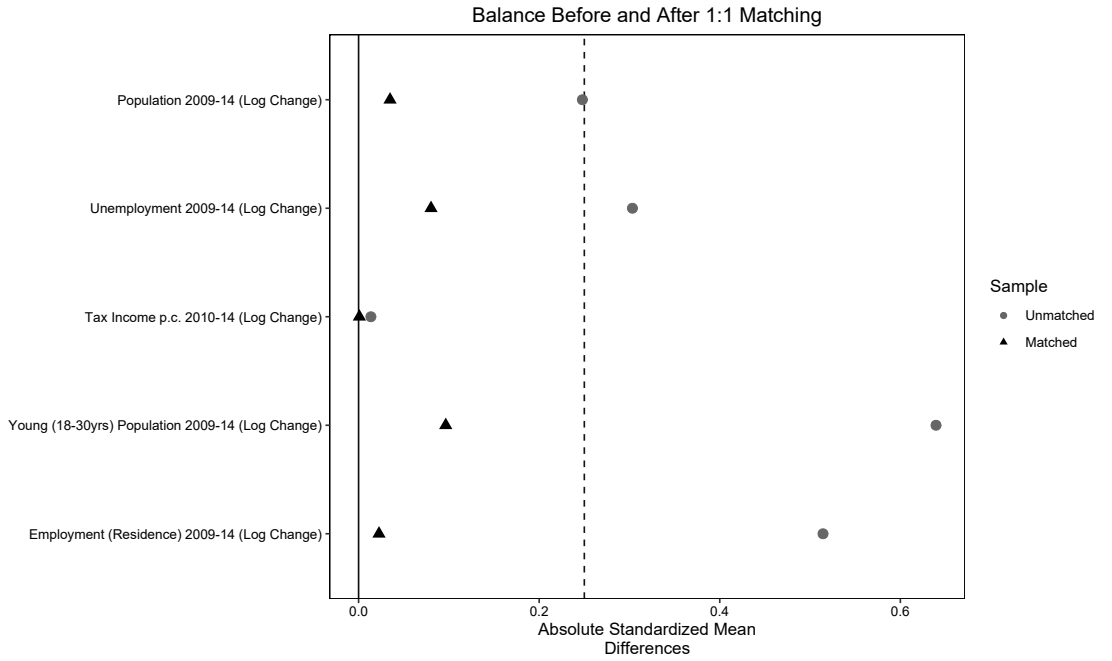


Figure 3. Balance plot showing standardized mean differences and variance ratios between treated municipalities and matched control municipalities outside Bavaria before and after Mahalanobis nearest-neighbor matching.

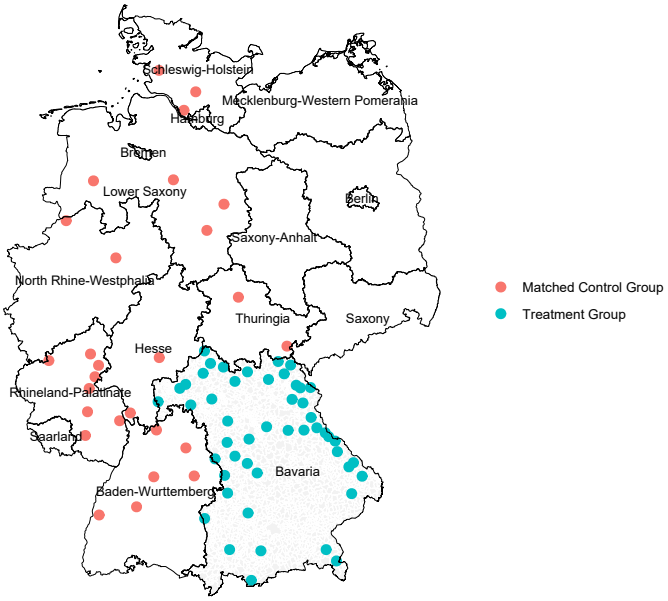


Figure 4. Map of Germany showing treated municipalities in Bavaria in teal and their matched control counterparts outside Bavaria in red.

Variable	Mean Treated	Mean Control	Std. Mean Diff.	Var. Ratio
Population	-0.01	-0.02	0.04	0.13
Unemployment	-0.25	-0.24	-0.10	0.14
Employment	0.10	0.07	0.83	0.08
Income per taxpayer	0.14	0.15	-0.12	0.23
Population 18–30 yrs	-0.03	-0.10	1.09	0.07

Table 7. Balance before matching between treated municipalities in Bavaria and other municipalities outside Bavaria.

Variable	Mean Treated	Mean Control	Std. Mean Diff.	Var. Ratio
Population	-0.01	-0.02	0.03	0.96
Unemployment	-0.25	-0.24	-0.07	1.02
Employment	0.10	0.09	0.10	0.94
Income per taxpayer	0.14	0.15	-0.04	1.08
Population 18–30 yrs	-0.03	-0.03	0.06	0.94

Table 8. Balance after matching between treated municipalities in Bavaria and matched municipalities outside Bavaria.

and Sant'Anna (2021) and Sun and Abraham (2021), relying on clean control groups of never-treated units.

Both estimators decompose the treatment effect by treatment cohort, defined by the year in which a municipality first receives a relocation. I then aggregate the cohort-specific effects to obtain standard event-study parameters. These aggregated effects can be averaged over calendar time or over event time.

For the Callaway & Sant'Anna estimator, the cohort-specific average treatment effect is defined as

$$ATT(g,t) = E\left[Y_{it}(g) - Y_{it}(\infty) \mid \mathcal{G}_i = g\right]. \tag{1}$$

The effect is identified by comparing the expected change in outcomes for cohort g between periods $g-1$ and t with the corresponding change for units that are either not-yet-treated or never treated:

$$ATT(g,t) = E\left[Y_{it} - Y_{i,g-1}(\infty) \mid \mathcal{G}_i = g\right] - E\left[Y_{it} - Y_{i,g-1}(\infty) \mid \mathcal{G}_i \in \mathcal{G}_{comp}\right], \tag{2}$$

where $\mathcal{G}_{comp} = \{g' : g' > t\}$. To aggregate these cohort-specific effects into a single event-study parameter, I use

$$ATT_l^w = \sum_g w_g \times ATT(g, g+l), \tag{3}$$

where l represents event time, measured as the number of periods after treatment. For statistical inference, I use bootstrap methods.

The Sun & Abraham estimator works similarly by modifying two-way fixed effects estimation and recovering cohort-specific average treatment effects using auxiliary regressions. The main difference is that, in the Callaway & Sant'Anna specification, the parallel trends assumption only needs to hold relative to one pre-treatment period per cohort. Therefore, all coefficients are calculated as long differences relative to the first pre-treatment period for each cohort. In the event-study graph below, I translate the results to a common base period, making the coefficients more comparable to the previous long-difference estimates.

5 Results

5.1 First Stage: Public Employment Relocation

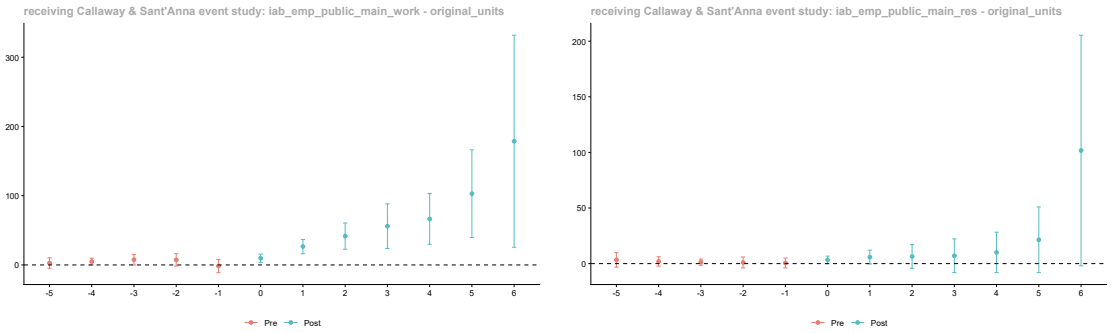
Before turning to the effects of the relocation program on local labor-market and population outcomes, I document the first stage of the policy using newly processed IAB employment data. The first-stage exercise asks whether treated receiving municipalities experienced an increase in public employment after the start of the relocation program. This is important because the main empirical analysis currently uses municipal employment data that do not yet separately identify civil servants and therefore subtracts relocated jobs as a conservative adjustment for the mechanical effect of the policy.

Fig. 5 shows event-study estimates for public employment in receiving municipalities. Panel A reports public employment by workplace, while Panel B reports public employment by place of residence. The workplace-based first stage increases sharply after treatment, confirming that the policy generated a discrete increase in public jobs at receiving workplaces. The residence-based first stage also increases, although less mechanically, because public employees may choose whether to relocate, commute, or remain in their original place of residence. The difference between the two panels is therefore informative about the spatial adjustment margin of public workers. Jobs are assigned to receiving workplaces by policy, while residence choices adjust endogenously.

These first-stage results support the interpretation of the reduced-form estimates as responses to actual public-employment relocations rather than only to policy announcements. They also motivate the distinction between workplace-based and residence-based employment throughout the empirical analysis and in the quantitative spatial model.

5.2 Long-Difference Results

Table 9 presents the main long-difference estimates. The table compares changes in unemployment and employment shares between 2014 and 2022, and changes in total population and working-age population between 2014 and 2023, for treated Bavarian municipalities and matched control municipalities outside Bavaria. Column 1 shows that the relocation program reduced the unemployment share in treated municipalities by approximately 7.9% relative to matched controls, although the estimate is not statistically significant at conventional levels. Column 2 shows that the employment share increased significantly by roughly 2% over the same period. Columns 3 and 4 show that total



(a) Change in public workplace employment

(b) Change in public residence employment

Figure 5. First-stage effects of the relocation program on public employment in receiving municipalities. The figure reports Callaway–Sant'Anna event-study estimates using IAB employment data. Panel A shows public employment by workplace. Panel B shows public employment by place of residence.

population and working-age population also increased relative to matched controls, although neither estimate is statistically significant in 2023.

	Unemp. Share	Emp. Share	Pop.	Working-Age Pop.
Treatment (Bin)	-0.0785	0.0201**	0.0067	0.0229
	(0.0564)	(0.0080)	(0.0111)	(0.0136)
Controls	✓	✓	✓	✓
Observations	76	76	76	76
R^2	0.73549	0.70676	0.72743	0.70365
Within R^2	0.19972	0.42486	0.15903	0.17938
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports long differences in unemployment and employment shares between 2014 and 2022, and log differences in total population and working-age population between 2014 and 2023. The control group consists of municipalities outside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 9. Main long-difference results for unemployment, employment, total population, and working-age population.

5.2.1 Labor-Market Outcomes

Table 10 presents long-difference estimates for employment shares for each endpoint year from 2015 to 2022, relative to 2014. The treatment effect in the first row represents the log difference in employment shares between treated Bavarian municipalities and matched untreated municipalities outside Bavaria. The estimated treatment effects are positive throughout, indicating that the program increased employment shares in treated municipalities relative to matched controls. The effect is largest in 2021 and is statistically significant from 2019 onward.

Table 11 presents the corresponding results for unemployment shares. In 2015, unemployment shares declined by 4.8%, although the effect is not statistically significant. In 2017 and 2018, unemployment shares fell significantly, by 5.6% and 11.8%, respectively. The positive and significant coefficient in 2016 is consistent with a short-run transition period. One possible explanation is an initial inflow of job seekers following the announcement of the relocation program. Another possibility is that relocating households temporarily increase local unemployment if other household members need time to find employment in the receiving municipality.

	2015	2016	2017	2018	2019	2020	2021	2022
Treatment (Bin)	0.0114	0.0109	0.0015	0.0069	0.0164*	0.0199**	0.0225**	0.0201**
	(0.0073)	(0.0096)	(0.0065)	(0.0067)	(0.0088)	(0.0076)	(0.0084)	(0.0080)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	75	76	76	76	76	76	76
R^2	0.64451	0.50807	0.42891	0.49072	0.50961	0.72563	0.71682	0.70676
Within R^2	0.18655	0.09062	0.12520	0.19133	0.24101	0.45450	0.50749	0.42486
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in employment shares for endpoint years from 2015 to 2022 relative to 2014. Column 1 shows the 2014–2015 log difference in employment share. Column 2 shows the 2014–2016 log difference in employment share, and so on. The control group consists of municipalities outside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 10. Log differences in employment shares between treated and untreated municipalities relative to 2014.

	2015	2016	2017	2018	2019	2020	2021	2022
Treatment (Bin)	-0.0483	0.0771***	-0.0556*	-0.1184**	-0.0682	-0.0672	-0.0906	-0.0785
	(0.0303)	(0.0259)	(0.0328)	(0.0437)	(0.0479)	(0.0485)	(0.0592)	(0.0564)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	75	76	76	76	76	76	76
R^2	0.59636	0.76802	0.65349	0.62658	0.65531	0.71969	0.70465	0.73549
Within R^2	0.17518	0.49257	0.13976	0.27633	0.27036	0.35671	0.34809	0.19972
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in unemployment shares for endpoint years from 2015 to 2022 relative to 2014. Column 1 shows the 2014–2015 log difference in unemployment share. Column 2 shows the 2014–2016 log difference in unemployment share, and so on. The control group consists of municipalities outside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 11. Log differences in unemployment shares between treated and untreated municipalities relative to 2014.

5.2.2 Population Outcomes

I repeat the long-difference analysis for total population and working-age population at the municipal level. As shown in Table 12, population started increasing slightly after treatment and reached a 1.3% increase in 2019. In 2017 and 2018, total population increased by 1.2% and 1.6%, respectively, in treated municipalities relative to matched controls. The estimates are statistically significant in both years.

When focusing on working-age population in Table 13, I find increases of 1.6% in 2017, 2.1% in 2018, and 1.8% in 2019. These results suggest that treated municipalities attracted new residents after relocations began, especially individuals with labor-market attachment. These residents may include movers from sending locations as well as other individuals attracted by new local employment opportunities. After 2019, the population response weakens, likely due to pandemic-related restrictions that made relocation more difficult. The expansion of remote work during the pandemic may also have reduced the need for employees to move to the receiving municipalities.

	2015	2016	2017	2018	2019	2020	2021	2022	2023
Treatment (Bin)	0.0012	0.0047	0.0119*	0.0156**	0.0127	0.0056	-0.0026	-0.0025	0.0067
	(0.0028)	(0.0048)	(0.0059)	(0.0075)	(0.0080)	(0.0079)	(0.0088)	(0.0101)	(0.0111)
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	76	76	76	76	76	76	76	76
R^2	0.61324	0.59471	0.65472	0.70977	0.68598	0.71950	0.72862	0.72492	0.72743
Within R^2	0.08832	0.08919	0.18766	0.23909	0.23355	0.18877	0.23241	0.19828	0.15903
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in total population for endpoint years from 2015 to 2023 relative to 2014. Column 1 shows the 2014–2015 log difference in total population. Column 2 shows the 2014–2016 log difference in total population, and so on. The control group consists of municipalities outside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 12. Log differences in population between treated and untreated municipalities relative to 2014.

	2015	2016	2017	2018	2019	2020	2021	2022	2023
Treatment (Bin)	0.0026	0.0078	0.0160*	0.0214**	0.0175*	0.0144	0.0045	0.0104	0.0229
	(0.0037)	(0.0064)	(0.0080)	(0.0088)	(0.0098)	(0.0087)	(0.0088)	(0.0118)	(0.0136)
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	75	76	76	76	76	76	76	76
R^2	0.59974	0.57962	0.62544	0.68325	0.67397	0.72029	0.73993	0.71406	0.70365
Within R^2	0.08166	0.10689	0.17604	0.26848	0.25232	0.23957	0.24922	0.21094	0.17938
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in working-age population for endpoint years from 2015 to 2023 relative to 2014. Column 1 shows the 2014–2015 log difference in working-age population. Column 2 shows the 2014–2016 log difference in working-age population, and so on. The control group consists of municipalities outside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 13. Log differences in working-age population between treated and untreated municipalities relative to 2014.

5.2.3 Sending Locations

Munich is the main sending location for the program, alongside seven other municipalities. I analyze the impact of the relocation program on these sending locations. Here, treatment is defined as the withdrawal of public-sector jobs at the municipal level. I therefore exclude new jobs created at receiving locations and top-ups at existing receiving agencies. Due to the small number of treated units, I use a simple two-period, static difference-in-differences specification to estimate the effect of public employment reallocation on sending locations. As in the main analysis, control municipalities are matched using Mahalanobis distance.

The employment share increases by 0.77 percentage points in sending locations, as shown in Table 14. The unemployment share decreases by 0.59 percentage points after public-sector employees relocate, as shown in Table 15. In terms of population outcomes, Table 16 shows that the working-age population share increases by 1.1 percentage points in sending locations. These results should be interpreted with caution because the treated municipalities vary substantially in size. For example, Munich, the third-largest city in Germany, is treated as one municipality alongside much smaller locations, such as Herrsching. Overall, the results suggest that relocations did not harm sending locations.

	Employment Share (Residence)		
Post (Binary) × Treatment (Binary) = 1	5.947***	0.7748**	0.7748**
	(0.7572)	(0.3800)	(0.3860)
Controls	✓		
Observations	192	192	192
R^2	0.19394	0.90383	0.90383
Within R^2		0.01321	0.01321
Municipality Fixed Effect		✓	✓
Year Fixed Effect		✓	✓
Mahalanobis Pair Fixed Effect			✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.
Notes: The table reports simple difference-in-differences estimates of the effect of public employment reallocation on residence-based employment shares in sending locations. The control group consists of municipalities outside Bavaria matched to sending municipalities in Bavaria using Mahalanobis distance.

Table 14. Two-period difference-in-differences results for the effect of public employment reallocation on employment in sending locations.

5.3 Event-Study Results

As a robustness check, I estimate event-study specifications using the Callaway and Sant'Anna (2021) estimator. This allows me to assess whether treatment-time heterogeneity affects the estimated treatment effects.

	Unemployment Share		
Post (Binary) × Treatment (Binary) = 1	-0.4625***	-0.5907***	-0.5907***
	(0.1369)	(0.1287)	(0.1307)
Controls	✓		
Observations	192	192	192
R^2	0.75983	0.92236	0.92236
Within R^2		0.12749	0.12749
Municipality Fixed Effect		✓	✓
Year Fixed Effect		✓	✓
Mahalanobis Pair Fixed Effect			✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.
Notes: The table reports simple difference-in-differences estimates of the effect of public employment reallocation on unemployment shares in sending locations. The control group consists of municipalities outside Bavaria matched to sending municipalities in Bavaria using Mahalanobis distance.

Table 15. Two-period difference-in-differences results for the effect of public employment reallocation on unemployment in sending locations.

5.3.1 Receiving Locations

Fig. 6 shows event-study estimates for two measures of employment shares. Panel A reports residence-based employment shares. Panel B reports the employment share for residents who also work in the treated municipality. Residence-based employment shares continue to decline in the first two years after treatment and then start to increase gradually. The largest estimate occurs seven years after treatment, suggesting that early treated cohorts may drive part of the effect. However, these estimates are not statistically significant. By contrast, the local employment share in Panel B increases immediately after treatment. The effect becomes statistically significant from the third post-treatment year onward.

Fig. 7 shows the corresponding event-study estimates for unemployment shares. Unemployment starts to decline after relocations begin in 2015.

Fig. 8 shows the event-study estimates for population outcomes. Total population starts to increase after treatment and peaks eight years after the policy begins. The increase in the working-age population share is more pronounced and becomes statistically significant six years after treatment.

Since the policy is ongoing and will be completed in 2025, these results remain preliminary. In the first years after implementation, the policy increased local employment shares, total population, and working-age population shares, while unemployment shares declined in receiving locations. The employment effects are especially pronounced for the local workforce, defined as residents who also work in the treated municipality. Together with the population increases, this pattern is consistent with relocated public employees moving from sending locations, with local residents taking up new jobs, or with additional

	Working-Age Population Share		
Post (Binary) × Treatment (Binary) = 1	-1.068**	1.111***	1.111***
	(0.4422)	(0.3210)	(0.3260)
Controls	✓		
Observations	192	192	192
R^2	0.67548	0.92396	0.92396
Within R^2		0.07336	0.07336
Municipality Fixed Effect		✓	✓
Year Fixed Effect		✓	✓
Mahalanobis Pair Fixed Effect			✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.
Notes: The table reports simple difference-in-differences estimates of the effect of public employment reallocation on working-age population shares in sending locations. The control group consists of municipalities outside Bavaria matched to sending municipalities in Bavaria using Mahalanobis distance.

Table 16. Two-period difference-in-differences results for the effect of public employment reallocation on working-age population shares in sending locations.

workers being attracted by improved local labor-market conditions.

5.3.2 Sending Locations

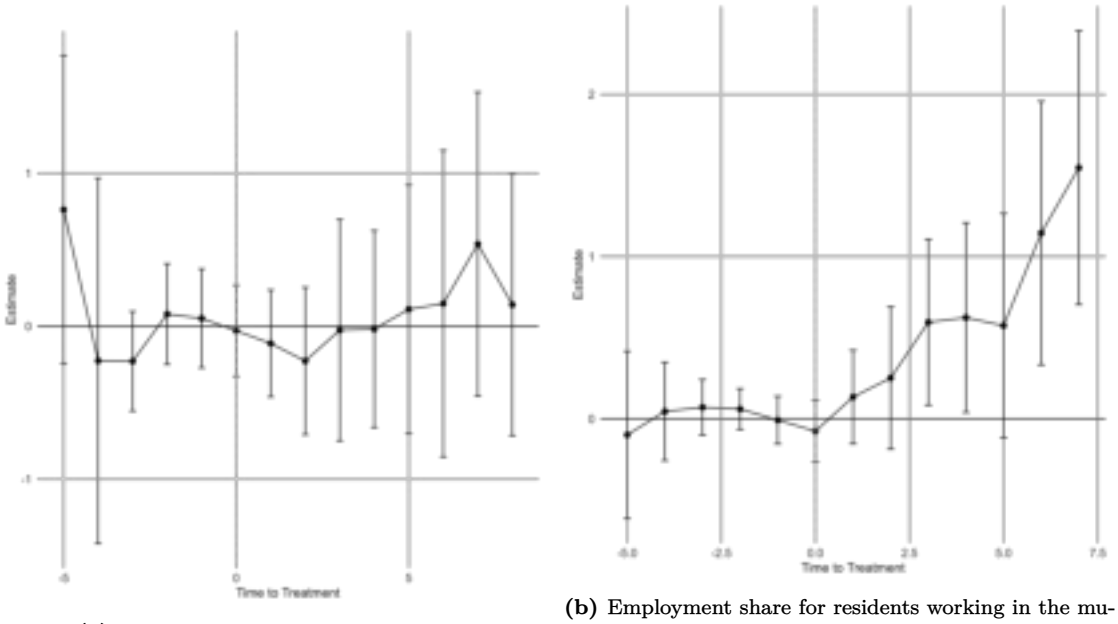
I define a sending municipality as treated in the first year it loses public jobs and estimate dynamic effects using the Callaway–Sant'Anna estimator with never-treated municipalities outside Bavaria as the comparison group. Event time is normalized so that $\tau = 0$ is the last pre-exposure year for each municipality, and I aggregate cohort-specific effects to event time.

Fig. 9 shows event-study estimates for employment shares in sending locations. Panel A reports residence-based employment shares. Panel B reports the employment share for residents who also work in the municipality. In both panels, post-exposure paths increase with treatment time up to seven years after relocation, indicating gains in employment shares at sending locations.

Fig. 10 shows no clear pre-trend and a decline in unemployment shares after exposure, consistent with reductions in unemployment in sending municipalities. The effect is largest, but also least precise, five years after the start of the relocation program.

For population outcomes, Fig. 11 shows increases in both total population and the working-age population share. Panel A shows that total population increases after the relocation program begins, with statistically significant estimates in the first post-treatment years. Panel B shows a clear upward trajectory in the working-age population share after treatment.

The sending-location outcome paths mirror the receiving-location event studies reported above. Employment shares increase, unemployment shares decrease, and working-age



(a) Residence-based employment share

(b) Employment share for residents working in the municipality

Figure 6. Callaway and Sant'Anna (2021) event-study estimates for employment shares in receiving locations. Treatment starts in 2015.

population shares rise at both ends of the relocation. This pattern is puzzling if the policy is interpreted only as subtracting public labor demand from origins and adding it to destinations. Sending locations do not contract. Instead, they appear to backfill the relocated public jobs and, if anything, improve after part of the public sector is reallocated elsewhere.

I interpret this symmetry as suggestive evidence of spillovers that operate both within and between sectors and that are not yet modeled in the existing public employment reallocation literature. To rationalize positive multipliers at destinations and non-negative dynamics at origins, I develop a quantitative spatial model in Section 8 that allows for within-sector and between-sector productivity spillovers, amenity spillovers, and spatial sorting.

5.4 Multiplier Analysis

I estimate a long-difference model at the municipality level to quantify the public-to-private employment multiplier. Let municipalities be indexed by m and employment outcome measures by d. I denote employment by $E_{m,d,t}$ in year $t \in \{2014, 2022\}$, and define the long difference as

$$\Delta E_{m,d} \equiv E_{m,d,2022} - E_{m,d,2014}. \tag{4}$$

I measure relocation intensity using two variables. The first is R_m^{all}, the total number of relocated public jobs and study places that municipality m receives between 2015 and

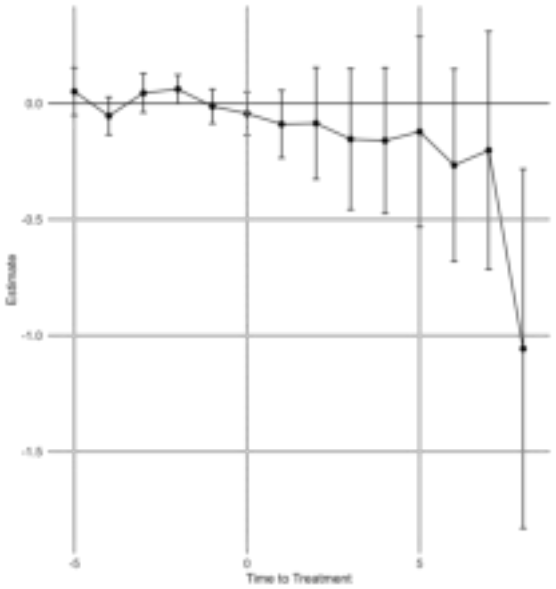


Figure 7. Callaway and Sant'Anna (2021) event-study estimates for unemployment shares in receiving locations. The unemployment share is defined as the number of unemployed persons in a municipality divided by the working-age population. Treatment starts in 2015.

2022. The second is R_m^{jobs}, the total number of relocated public jobs only. To remove the direct mechanical addition of relocated positions from the employment outcome, I define net employment changes as

$$\Delta E_{m,d}^{\text{net,all}} \equiv \Delta E_{m,d} - R_m^{\text{all}}, \tag{5}$$

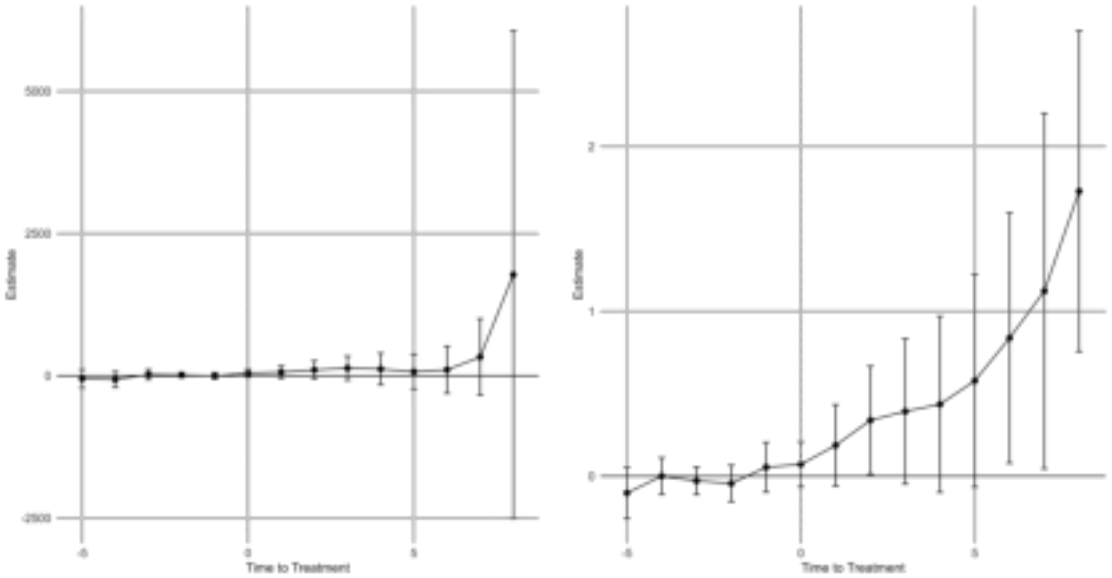
$$\Delta E_{m,d}^{\text{net,jobs}} \equiv \Delta E_{m,d} - R_m^{\text{jobs}}. \tag{6}$$

I control for pre-treatment characteristics X_m, including log changes in population, unemployment, tax income per capita, young population, and employment over the relevant pre-treatment periods, 2009–2014 or 2010–2014. Standard errors are clustered at the district level. I estimate

$$\Delta E_{m,d}^{\text{net,all}} = \alpha_d + \beta_d^{\text{all}} R_m^{\text{all}} + X_m' \gamma_d + \varepsilon_{m,d}, \tag{7}$$

$$\Delta E_{m,d}^{\text{net,jobs}} = \alpha_d^{\text{jobs}} + \beta_d^{\text{jobs}} R_m^{\text{jobs}} + X_m' \gamma_d^{\text{jobs}} + u_{m,d}. \tag{8}$$

I interpret β_d^{all} as the public-to-private jobs multiplier for outcome d, defined as net additional private jobs per relocated public job or study place. I interpret β_d^{jobs} analogously when relocation intensity is restricted to public jobs only. Fig. 12 relates R_m^{all} on the horizontal axis to $\Delta E_{m,d}^{\text{net,all}}$ on the vertical axis. The upward-sloping fitted line indicates that larger relocations are associated with larger net employment gains, consistent with a positive multiplier. For the employment measure where place of residence equals place of work, the median public-to-private employment multiplier is 1.08. This means that relocating 10 public jobs is associated with 10.8 additional local private jobs.



(a) Total population **(b)** Working-age population share

Figure 8. Callaway and Sant'Anna (2021) event-study estimates for population and working-age population shares in receiving locations.

To study heterogeneity, I construct municipality-level multipliers using the matched subclasses from the Mahalanobis matching procedure. Let $\widehat{Y}^{\text{all}}_{m,d}$ and $\widehat{Y}^{\text{jobs}}_{m,d}$ denote fitted values from the long-difference models above. I compare each treated municipality m to the predicted outcome of its matched control municipality c within the same Mahalanobis subclass:

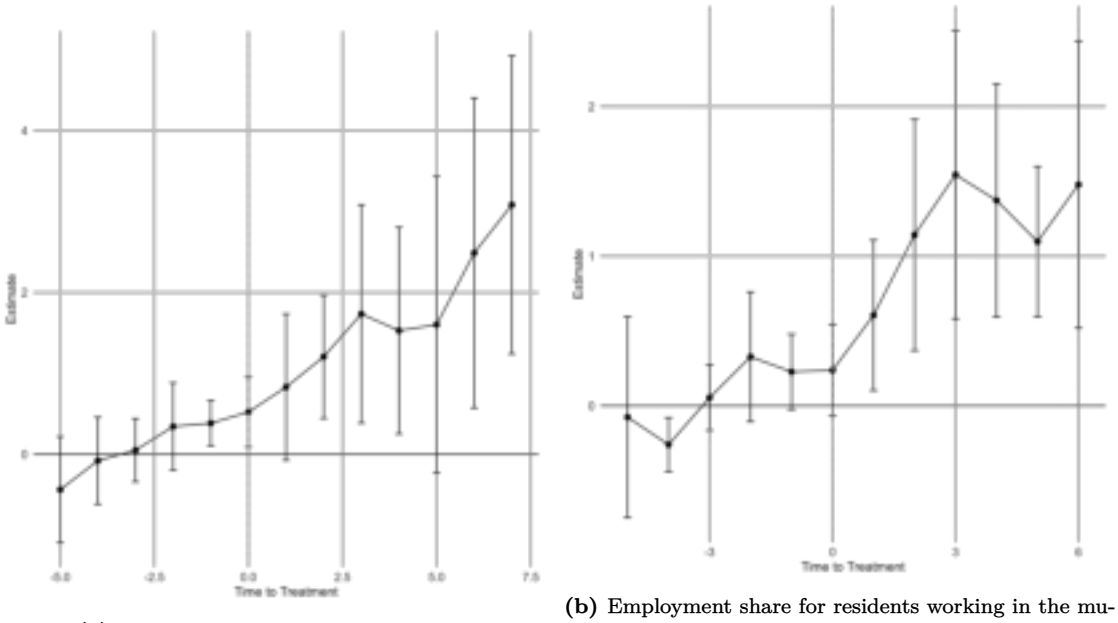
$$\widehat{\text{effect}}^{\text{all}}_{m,d} \equiv \widehat{Y}^{\text{all}}_{m,d} - \widehat{Y}^{\text{all}}_{c,d}, \tag{9}$$

$$\widehat{\mu}^{\text{all}}_{m,d} \equiv \frac{\widehat{\text{effect}}^{\text{all}}_{m,d}}{R^{\text{all}}_{m}}, \tag{10}$$

$$\widehat{\text{effect}}^{\text{jobs}}_{m,d} \equiv \widehat{Y}^{\text{jobs}}_{m,d} - \widehat{Y}^{\text{jobs}}_{c,d}, \tag{11}$$

$$\widehat{\mu}^{\text{jobs}}_{m,d} \equiv \frac{\widehat{\text{effect}}^{\text{jobs}}_{m,d}}{R^{\text{jobs}}_{m}}. \tag{12}$$

Fig. 13 maps $\widehat{\mu}^{\text{all}}_{m,d}$, measured as net private jobs per relocated public job or study place. The map shows where employment gains are concentrated across receiving municipalities.



(a) Residence-based employment share

(b) Employment share for residents working in the municipality

Figure 9. Callaway and Sant'Anna (2021) event-study estimates for employment shares in sending locations. Treatment starts in 2015.

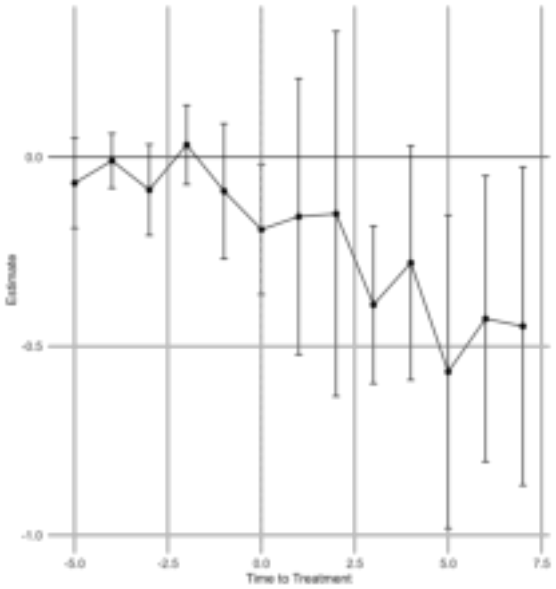
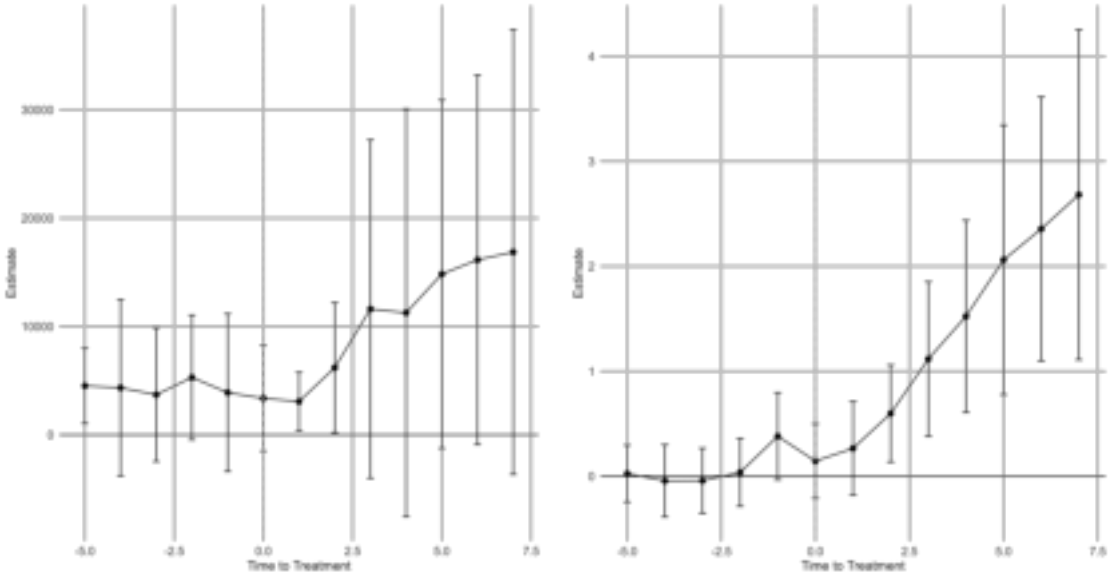


Figure 10. Callaway and Sant'Anna (2021) event-study estimates for unemployment shares in sending locations. The unemployment share is defined as the number of unemployed persons in a municipality divided by the working-age population. Treatment starts in 2015.



(a) Total population **(b)** Working-age population share

Figure 11. Callaway and Sant'Anna (2021) event-study estimates for population and working-age population shares in sending locations.

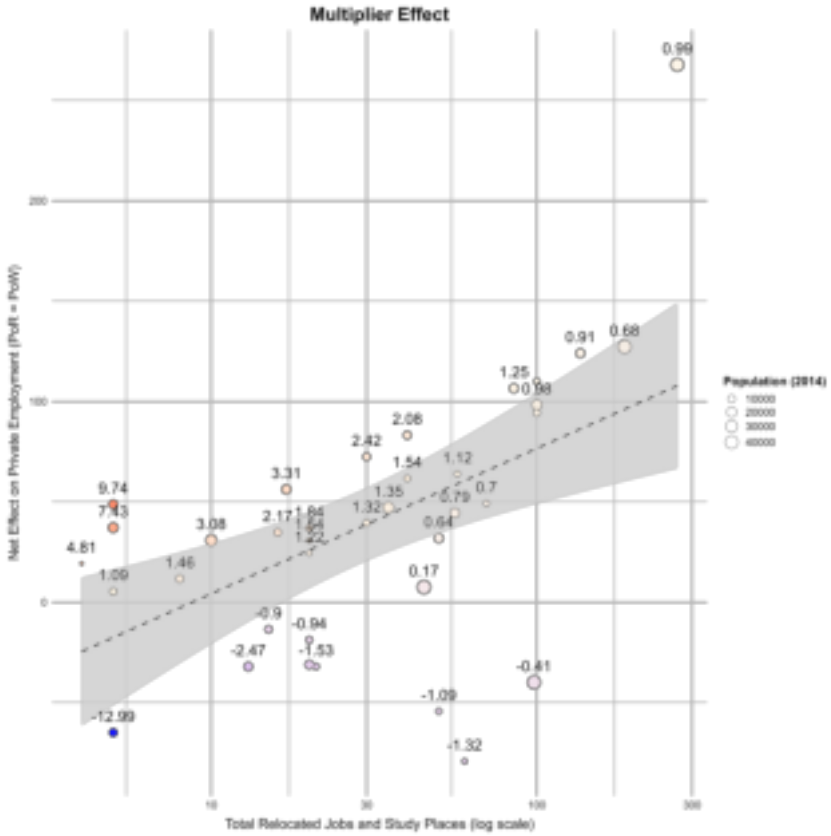


Figure 12. Relocation intensity and net employment gains for employment where place of residence equals place of work.

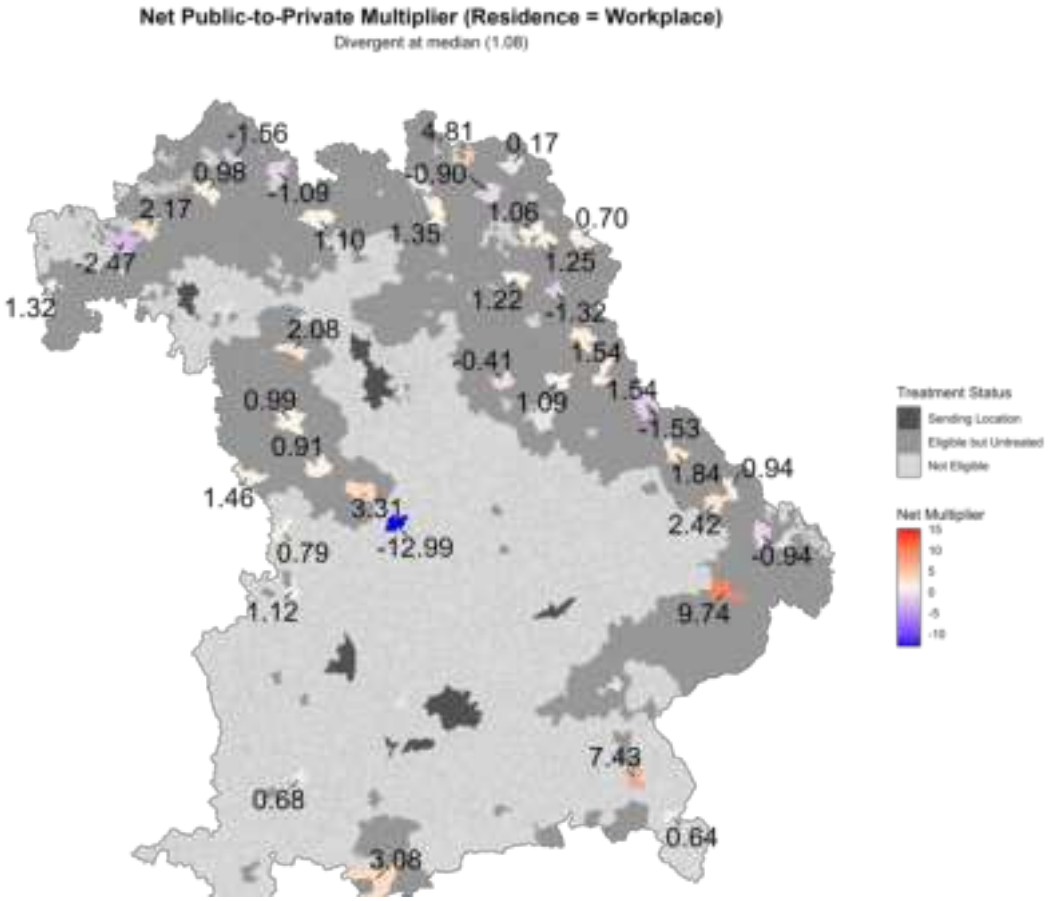


Figure 13. Municipality-level public-to-private employment multipliers above and below the state median.

6 Two-Sector Quantitative Spatial Model

The reduced-form results point to a mechanism that goes beyond the standard local-demand channel. Employment increases not only in receiving municipalities but also in sending locations. This pattern is difficult to rationalize if public-sector relocations are interpreted only as subtracting labor demand from origins and adding it to destinations. I therefore build a quantitative spatial model to discipline a richer general-equilibrium mechanism.

The model captures two forces. First, it captures a spatial misallocation logic. Public-sector jobs in this setting provide traded administrative services, so their baseline productivity depends less on local conditions than private-sector productivity does. Private-sector productivity, by contrast, varies strongly across space. In high-productivity urban sending locations such as Munich, the opportunity cost of public employment is high because public jobs occupy labor and space that could otherwise be used by highly productive private activity. In receiving locations, where private-sector productivity is lower, this opportunity cost is smaller. Reallocating a fixed number of public jobs from the former to the latter can therefore improve the spatial allocation of economic activity without increasing aggregate public employment.

Second, the model allows private-sector productivity to depend on sectoral spillovers. Within-private-sector spillovers can be stronger than public-to-private spillovers. As the policy changes local public-employment shares, receiving locations may benefit from the direct employment and local-demand shock, while sending locations may gain when labor and scarce urban space are reallocated toward private activity with stronger agglomeration externalities. I implement the *Heimatstrategie* as a counterfactual reallocation of public workplace employment and evaluate equilibrium impacts on private employment, productivity, wages, amenities, and rents. In ongoing work, I estimate the model's agglomeration spillover elasticities via GMM, targeting the reduced-form policy responses as moment conditions.

6.1 Model Setup

I adapt the quantitative spatial model of Ahlfeldt, Redding, et al. (2015) to study the general-equilibrium effects of reallocating state public employment within Germany. The economy is a closed German economy with exogenous total employment H. Space consists of discrete locations indexed by i for residence and j for workplace. In the current implementation, locations are counties, each endowed with K_i units of land. Workers are mobile across counties and choose residence–workplace pairs.

The economy has two sectors $m \in \{P, G\}$, where P denotes the private sector and G denotes the state public sector. Public workplace employment is exogenous and spatially allocated by policy. Private workplace employment is endogenous. Wages are not sector-specific. There is a single equilibrium wage w_j at workplace j, and workers in both sectors face the same commuting and residence choices.

Public sector. Public workplace employment is assigned to workplaces by policy. Let $\bar{H}^M_{G,j}$ denote public employment located at workplace j. The policy counterfactual changes the spatial allocation of $\{\bar{H}^M_{G,j}\}$, while holding aggregate public employment fixed. Public workers still choose where to live and commute, so public residence employment $H^R_{G,i}$ is endogenous through commuting probabilities.

Private sector. Private workplace employment $H^M_{P,j}$ is endogenous. The private sector is perfectly competitive, and wages and allocations adjust to clear markets given the exogenous spatial allocation of public employment. Total workplace employment at j is

$$H^M_j = H^M_{P,j} + \bar{H}^M_{G,j},$$

and total residents in i are

$$H^R_i = H^R_{P,i} + H^R_{G,i}.$$

6.1.1 Workers

A worker of type o choosing residence i and workplace j obtains indirect utility

$$u_{ijo} = \frac{z_{ijo} B_i w_j Q_i^{\beta-1}}{d_{ij}}, \tag{13}$$

where B_i denotes residential amenities, w_j is the workplace wage, Q_i is the housing price, and d_{ij} captures commuting costs. Commuting disutility increases in travel time according to

$$d_{ij} = \exp(\kappa \tau_{ij}),$$

where τ_{ij} is road travel time and $\kappa > 0$ is the commuting-cost parameter. Idiosyncratic preferences z_{ijo} are Fréchet distributed, which implies gravity-style commuting flows. The shape parameter ϵ governs taste dispersion. A higher ϵ implies that workers are more responsive to differences in wages, amenities, housing costs, and commuting costs.

6.1.2 Private Firms

Private firms produce a freely traded numeraire good under perfect competition. Production in workplace j uses private labor and commercial floor space:

$$y_j = A^P_j \left(H^M_{P,j}\right)^{\alpha} (L_{M,j})^{1-\alpha}, \tag{14}$$

where $H^M_{P,j}$ denotes private workplace employment, $L_{M,j}$ is commercial floor space, $\alpha \in (0,1)$ is the labor share, and A^P_j is private-sector productivity. Firms take wages, commercial rents, and productivity as given. Perfect competition and constant returns to scale imply zero profits in equilibrium.

6.1.3 Within- and Between-Sector Productivity Spillovers

Private-sector productivity is shaped by sector-specific workplace density spillovers:

$$A_j^P = a_j \left(\Upsilon_j^P\right)^{\lambda} \left(\Upsilon_j^G\right)^{\sigma\lambda}, \tag{15}$$

where a_j is an exogenous productivity fundamental, λ governs within-private-sector spillovers, and σ scales the strength of public-to-private spillovers relative to within-private-sector spillovers. I define sector-specific workplace density as

$$\Upsilon_j^m = \sum_s \exp(-\delta\tau_{js}) \frac{H_{m,s}^M}{K_s}, \qquad m \in \{P, G\}, \tag{16}$$

where δ controls spatial decay. In the counterfactual, public workplace employment $\bar{H}_{G,s}^M$ is shocked exogenously. This affects A_j^P directly through the public density term Υ_j^G and indirectly through the general-equilibrium response of private workplace employment $H_{P,s}^M$.

The key model restriction is that within-private-sector spillovers may be stronger than public-to-private spillovers. This formalizes the mechanism in the introduction. Removing public employment from high-productivity urban origins can increase the relative importance of private activity in places where private-to-private spillovers are strongest, while adding public jobs to receiving locations can generate local demand and employment effects where the opportunity cost of public employment is lower.

6.1.4 Amenities, Construction, and Market Clearing

Residential amenities are an endogenous function of surrounding resident density:

$$B_i = b_i \Omega_i^{\eta}, \qquad \Omega_i = \sum_s \exp(-\rho\tau_{is}) \frac{H_s^R}{K_s}, \tag{17}$$

where b_i is an exogenous amenity fundamental, η governs the strength of amenity spillovers, and ρ controls spatial decay. Amenities are not sector-specific. The public sector affects amenities only through general-equilibrium population reallocation.

The construction sector follows Ahlfeldt, Redding, et al. (2015). Firms combine land and capital to produce floor space, and floor space within each location is allocated between residential and commercial use. Commuting markets, land markets, and goods markets clear in equilibrium. Total employment H is fixed at the country level, while equilibrium utility is endogenous.

6.2 Quantification

I quantify the model at the county level. The current version follows the structure of Ahlfeldt, Redding, et al. (ibid.), adds an exogenous public sector, and introduces asymmetric productivity spillovers between private and public employment.

Parameters. The current calibration uses standard parameter values from the urban and spatial literature. The commercial floor-space share in production is $1-\alpha = 0.20$, the expenditure share on housing is $1-\beta = 0.25$, and the share of land in property value is $1-\mu = 0.25$. Other parameters follow Ahlfeldt, Redding, et al. (2015). I estimate the commuting parameters using commuting flows and travel times. At the county level, the estimates are

$$\widehat{\epsilon} = 3.11, \qquad \widehat{\kappa} = 0.0034.$$

The spillover parameters are currently untargeted. In the simulations below, I set $\lambda = 1$, $1-\sigma = 0.3$, and $\xi = 1$, so that $Q_i = Q_{i,\text{obs}}$. In ongoing work, I will estimate the spillover elasticities by targeting the empirical policy responses.

6.3 Policy Counterfactual

Let $\bar{H}_{Mj}^{G,0}$ denote baseline public workplace employment in location j. The counterfactual reallocates public jobs across space according to the *Heimatstrategie*:

$$\bar{H}_{Mj}^{G,\text{cf}} = \bar{H}_{Mj}^{G,0} + \Delta H_{Mj}^{G}, \tag{18}$$

$$\sum_j \Delta H_{Mj}^{G} = 0. \tag{19}$$

Total public employment is unchanged. The policy only changes its spatial allocation. Public jobs are assigned to workplaces by policy, but public workers choose where to live and commute. Hence public residence employment H_{Ri}^{G} is endogenous.

In the counterfactual, I re-solve for wages, housing prices, commuting flows, private employment, public residence employment, productivity, and amenities. The central endogenous outcomes are

$$\left(w_j^{\text{cf}}, Q_i^{\text{cf}}, H_{P,j}^{M,\text{cf}}, H_{P,i}^{R,\text{cf}}, H_{G,i}^{R,\text{cf}}, A_j^{P,\text{cf}}, B_i^{\text{cf}}\right).$$

The resulting equilibrium compares the baseline spatial allocation to a counterfactual in which the same aggregate number of public jobs is redistributed across locations.

6.4 Counterfactual Results

This section reports the current county-level simulation results. Each figure plots the percentage change from the baseline equilibrium to the counterfactual equilibrium induced by the public-employment reallocation.

6.4.1 Private Workplace Employment

Fig. 14 shows that the counterfactual raises private workplace employment in most Bavarian counties, including both sending and receiving locations. The gains are strongest near metropolitan areas and weaker in more rural parts of Bavaria. Declines are concentrated mainly outside Bavaria. This crowding-in pattern is the main quantitative implication of the model. It rationalizes the reduced-form finding that employment can increase at

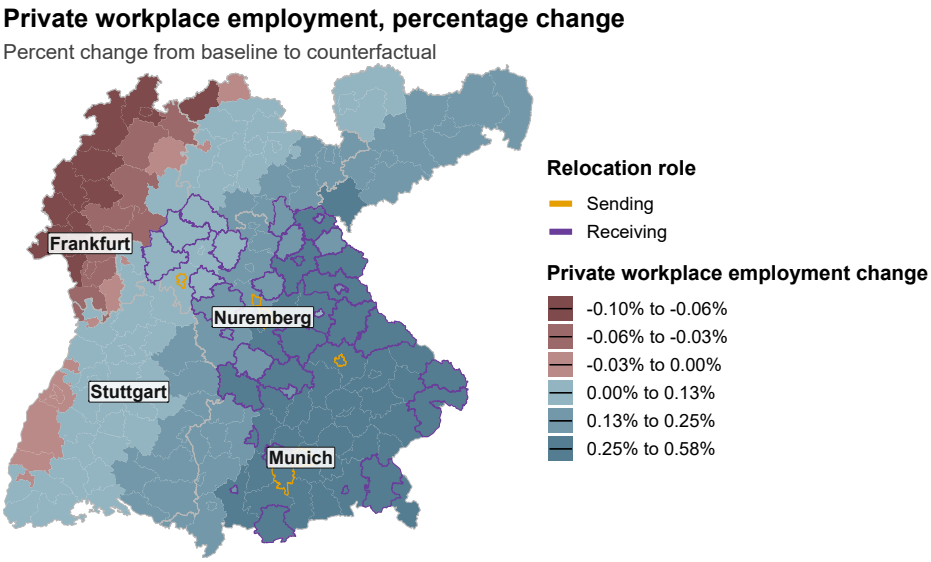


Figure 14. Counterfactual change in private workplace employment $H^M_{P,j}$.

both ends of the relocation: receiving locations gain from the public-employment shock, while sending locations gain as labor and space are reallocated toward private activity with stronger agglomeration externalities.

6.4.2 Productivity

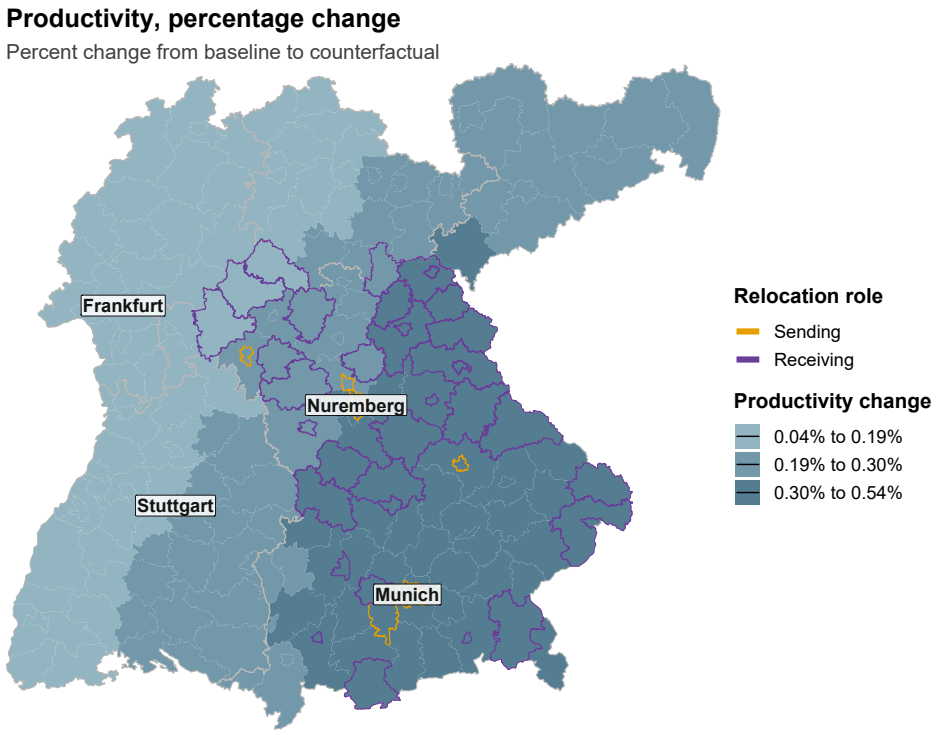


Figure 15. Counterfactual change in private-sector productivity A_j^P.

Fig. 15 shows that model-implied private-sector productivity increases in both sending and receiving counties. The gains are especially strong in counties close to Munich. At baseline, sending locations tend to be among the most productive counties, while receiving locations tend to be among the least productive. The counterfactual therefore reflects both the direct change in public workplace density and the endogenous response of private workplace employment through within-private-sector spillovers.

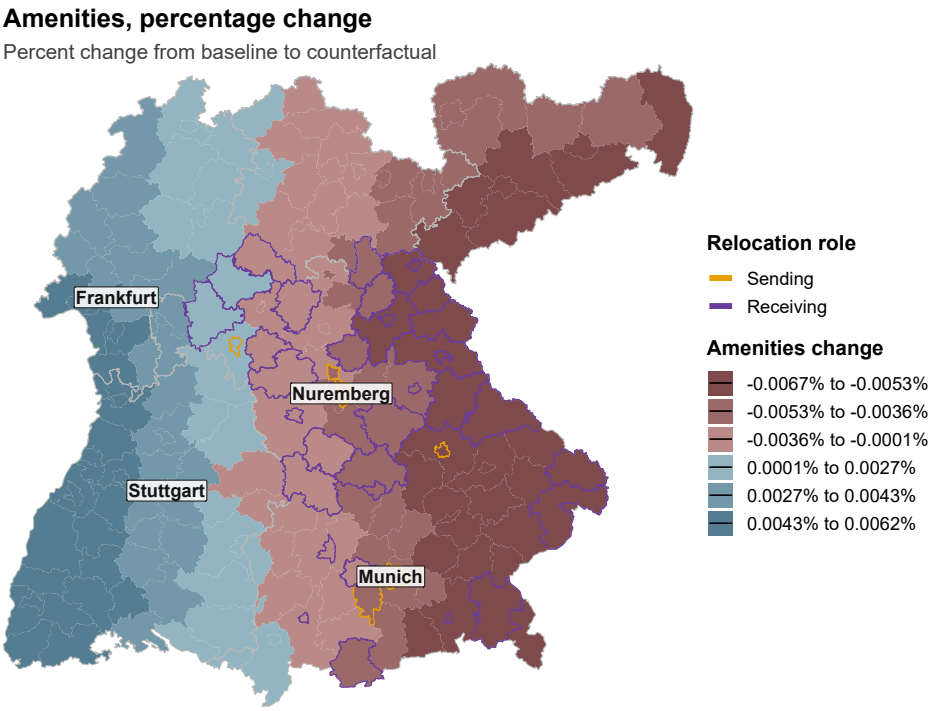


Figure 16. Counterfactual change in residential amenities B_i.

6.4.3 Amenities

Fig. 16 shows that amenities decline in most Bavarian counties under the current calibration. This suggests that congestion and residential general-equilibrium forces dominate amenity gains in the present version of the model. Amenity increases are concentrated mostly outside Bavaria. This result should be interpreted as preliminary because the current amenity specification depends on total resident density and does not yet incorporate richer local public-good or fiscal channels.

6.4.4 Wages

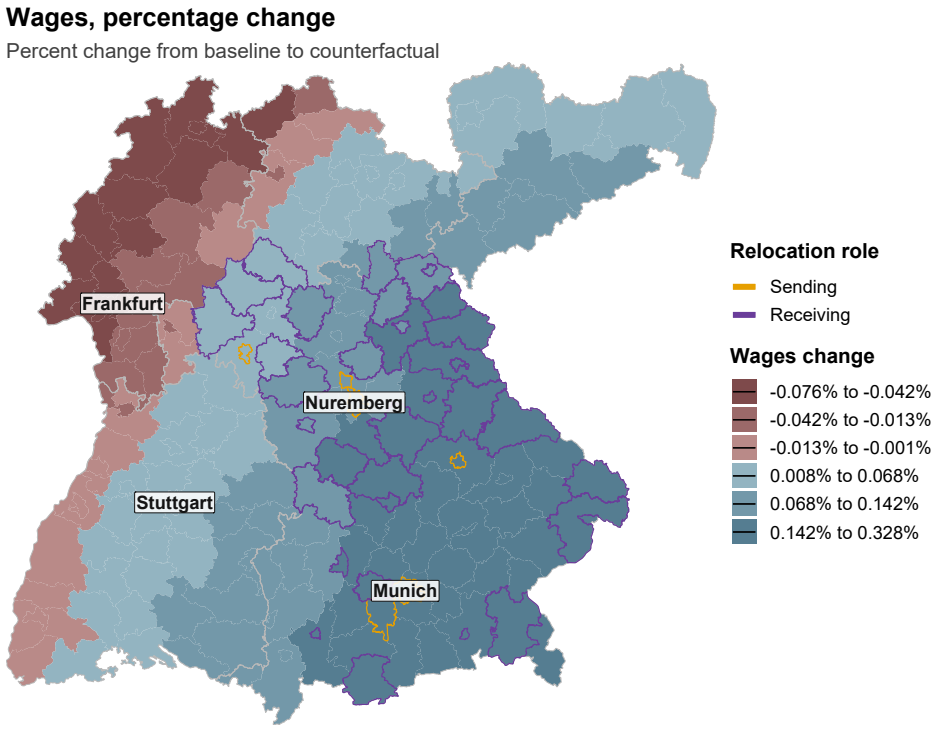


Figure 17. Counterfactual change in wages w_j.

Fig. 17 shows that wages increase within Bavaria, including in both sending and receiving locations, while wage declines are concentrated mainly outside Bavaria. This pattern reflects the equilibrium adjustment of private employment, commuting flows, and productivity after the public-employment reallocation.

6.4.5 Total Workplace Employment

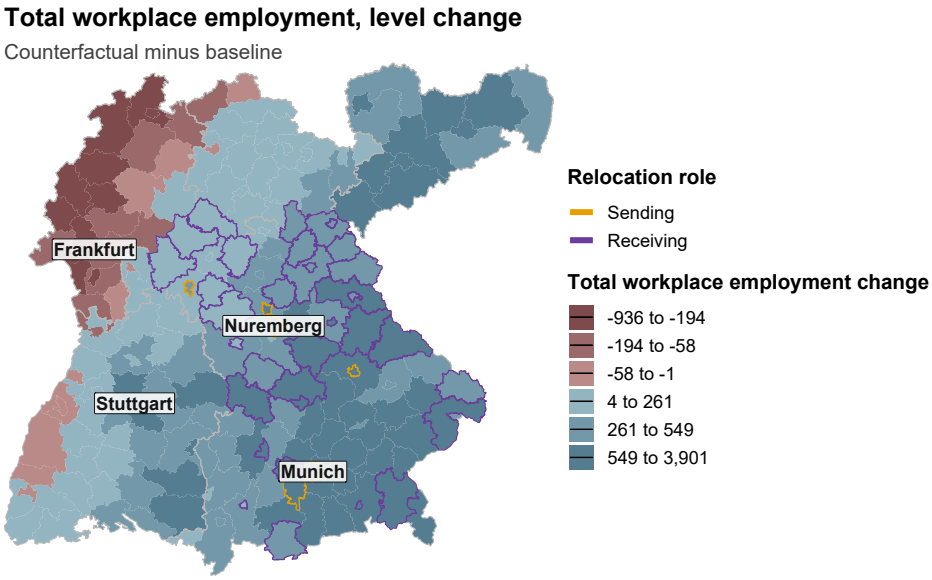


Figure 18. Counterfactual change in total workplace employment H_j^M.

Fig. 18 reports changes in total workplace employment. Since the policy reallocates public workplace employment mechanically while private employment adjusts endogenously, this outcome combines the direct public-employment shock with the induced private-sector response.

6.4.6 Residence Employment

Fig. 19 shows how the counterfactual changes the spatial distribution of resident employment. Because workers choose residence and workplace pairs endogenously, residence employment need not move one-for-one with the policy-induced change in public workplace employment.

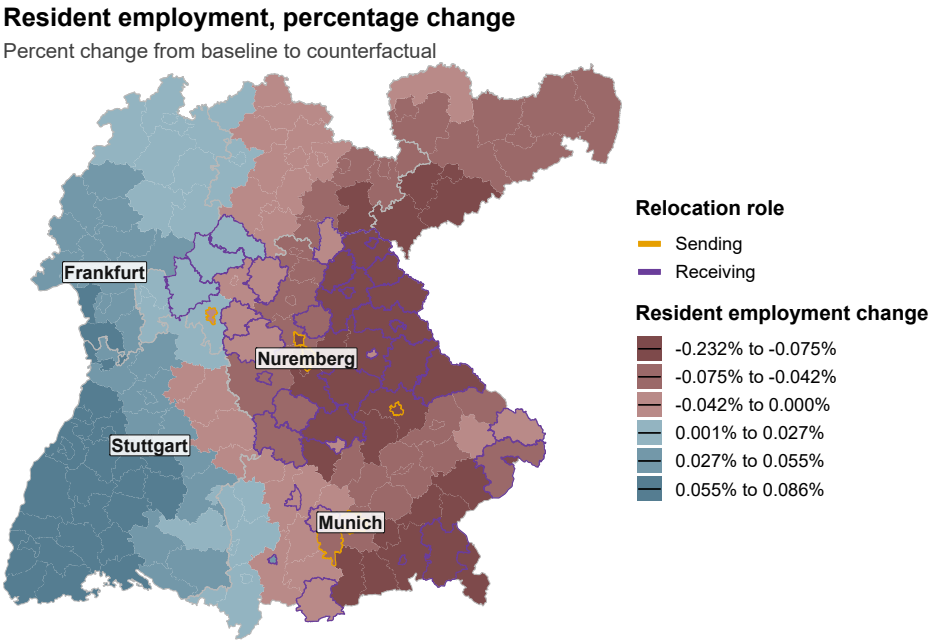


Figure 19. Counterfactual change in residence employment H_i^R.

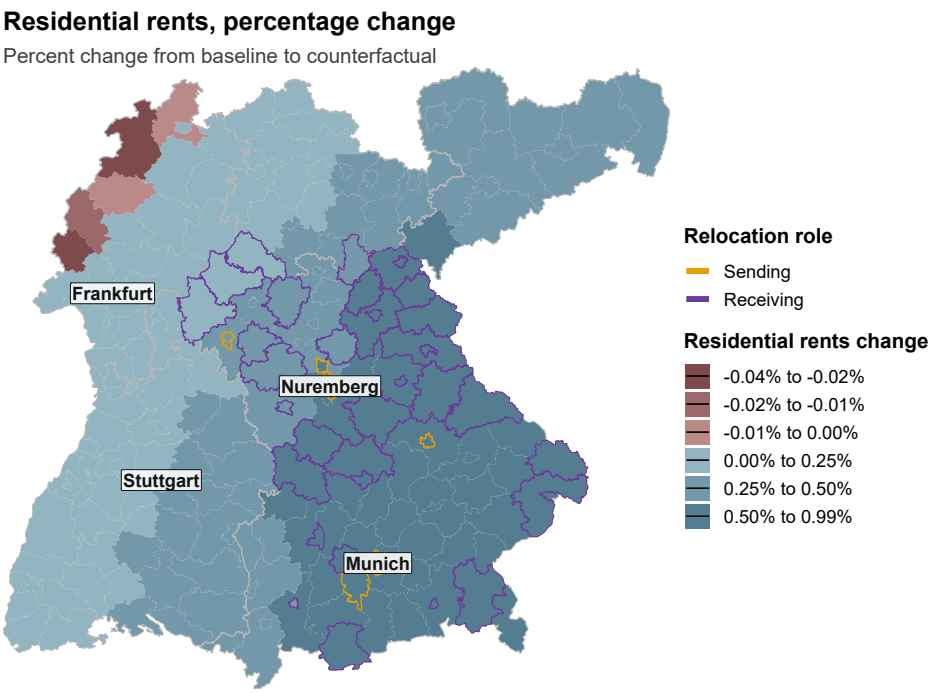


Figure 20. Counterfactual change in residential rents $Q_{\text{res},i}$.

6.4.7 Residential Rents

Fig. 20 reports changes in residential rents. This margin is relevant because public-sector relocations affect local labor demand and residential sorting. In future versions, I will extend this part of the model to incorporate the commercial floor-space margin more explicitly, including the release of office floor space in prime urban sending locations.

7 Conclusion

This paper studies whether relocating public-sector employment can operate as an effective place-based policy. I examine Bavaria's *Heimatstrategie*, a policy that reallocates public-sector jobs and apprenticeships from urban sending locations, especially Munich, toward smaller and economically lagging municipalities. The setting is useful for studying public employment reallocation because treatment assignment follows an explicit policy allocation rule, implementation occurs across many municipalities, and the policy affects both receiving and sending locations.

The empirical results show that public employment reallocation improves labor-market and demographic outcomes in receiving municipalities. After netting out the mechanical addition of relocated public jobs, employment shares increase by up to 2.3% or 1.3 percentage points by 2021. Unemployment falls significantly in 2017 and 2018, with the largest estimated decline reaching 11.9% or 0.33 percentage points in 2018 relative to 2014. Total population and the working-age population also increase, with the working-age population share rising by up to 2.1% in 2018. The estimates imply a public-to-private jobs multiplier of 1.08 for employment where place of residence equals place of work.

A central finding is that sending locations are not harmed by the relocation policy. Instead, employment shares increase, unemployment falls, and the working-age population share rises in sending municipalities. This pattern is difficult to reconcile with a simple local-demand mechanism in which removing public jobs from origin locations mechanically reduces local private activity. It suggests that public employment reallocation can create adjustment margins at both ends of the policy: receiving locations benefit from the employment and demand shock, while sending locations may benefit when labor and scarce urban space are reallocated toward private activity.

To rationalize these patterns, I build a two-sector quantitative spatial model with a private sector and a state public sector. The model formalizes two mechanisms. First, public-sector jobs in this setting provide traded administrative services, so their baseline productivity depends less on local conditions than private-sector productivity does. Reallocating public jobs from high-productivity urban origins to lower-productivity receiving locations can therefore reduce the opportunity cost of public employment. Second, private-sector productivity depends on sectoral spillovers, with within-private-sector spillovers potentially stronger than public-to-private spillovers. Under the current county-level calibration, the counterfactual relocation increases private employment, productivity, and wages across much of Bavaria, including both sending and receiving counties. Amenities decline in most Bavarian counties under the current calibration, suggesting that congestion and residential general-equilibrium forces dominate amenity

gains in this version of the model.

The paper contributes to the literature on place-based policies, public employment reallocation, and spatial equilibrium models with public employment. It studies public employment reallocation as a distinct place-based policy instrument, exploits the policy allocation rule to address selection into treatment, and estimates effects for both receiving and sending locations. It also provides a framework for rationalizing why reallocating a fixed number of public jobs need not be zero-sum across space.

Several components remain work in progress. First, I am constructing a yearly municipal employment panel for Germany that combines the universe of social-security records from the IAB Integrated Employment Biographies with the universe of civil servants by workplace and place of residence. This will allow me to measure total local employment and to obtain a finer sectoral breakdown than is available in the German Regional Statistical Database. Second, I will extend the model and empirical data with commercial floor-space and commercial price information based on the new Ahlfeldt, Heblich, Seidel, and Yin data. This will allow me to model more explicitly the real-estate margin of the policy, including the release of office floor space in prime urban sending locations. Third, I will estimate the model's agglomeration spillover elasticities via GMM, targeting the empirical policy responses as moment conditions, and then re-simulate the model at the municipal level.

Appendix

Appendix A: Institutional Setting

Structured News Search

I conducted a structured German-language news search in Nexis Uni using the search string

Behördenverlagerung OR Behördenumzug OR "Umzug von Behörden" OR "Umzug vom A

I restricted the search period to articles published from 1991 onward and limited the sample to news articles, excluding private-company publications, individual webpages, and legal texts.

The query yielded 1,531 results. I identified news reports on 22 federal and 87 state agencies between 2005 and 2032. In total, the reports cover 13,395 relocated jobs. The largest number of relocations occurs in 2019, with 5,843 jobs, followed by 2030, with 2,300 jobs, and 2021, with 1,070 jobs. Table 17 reports the number of relocated jobs and study places by destination state, and Table 18 lists all sending and destination cities.

The search identifies both single-agency relocations and broader relocation programs. The three main programs are the federal program for coal regions, the Bavarian *Heimatstrategie*, and the Saxon *Lausitzprogramm 2038*. I use this search to document the broader institutional context of public employment reallocation in Germany.

State	N of Relocated Jobs	N of Relocated Study Places
Bavaria	9,681	1,730
Brandenburg	510	
Hesse	100	
Saxony	2,214	
Saxony-Anhalt	850	
Thuringia	40	

Table 17. Number of public-sector jobs and study places relocated to each state according to the Nexis Uni news search.

Sending Cities	Destination Cities
Ammersee	Amberg, Ansbach, Augsburg
Berlin, Bonn	Bad Kissingen, Bad Königshofen im Grabfeld,
Bad Kissingen	Bad Kötzting, Bad Neustadt an der Saale, Bad Reichenhall,
	Bad Steben, Bautzen, Berlin, Bischofswerda, Borna
	Cottbus
Dresden	Dessau, Dinkelsbühl, Dresden
Erlangen, Eschborn, Essen	Ebern, Eichstätt
	Freital, Freyung, Fürstenfeldbruck, Fürth
Großenhain	Garmisch-Partenkirchen, Gemünden am Main, Gera,
	Görlitz, Grafenau, Gunzenhausen
Herrsching a. Ammersee	Halle (Saale), Höchstädt a.d.Donau, Hof
	Ingolstadt
	Kaufbeuren, Kemnath, Kitzingen, Kronach, Kulmbach
	Lauterbach, Leipzig, Lohr am Main
München	Marktredwitz, Meißen, Merseburg, Mittenwald
	Mühldorf a. Inn, Münchberg
Nürnberg	Nabburg, Neustadt an der Aisch, Nördlingen, Nürnberg
	Obernburg am Main, Oberviechtach
	Pegnitz
Regensburg	Regen, Rosenheim
Starnberg	Schwandorf, Schweinfurt
	Tirschenreuth, Traunstein
	Vilshofen an der Donau, Vohenstrauß
Würzburg	Waldkirchen, Waldmünchen, Waldsassen, Weiden in der Oberpfalz
	Weißenburg i. Bay., Weißwasser, Windischeschenbach, Wunsiedel
	Zittau, Zwiesel

Table 18. Sending and destination cities identified in the Nexis Uni news search, listed alphabetically. Some cities, such as Berlin and Dresden, appear both as sending and destination cities.

Appendix B: Data

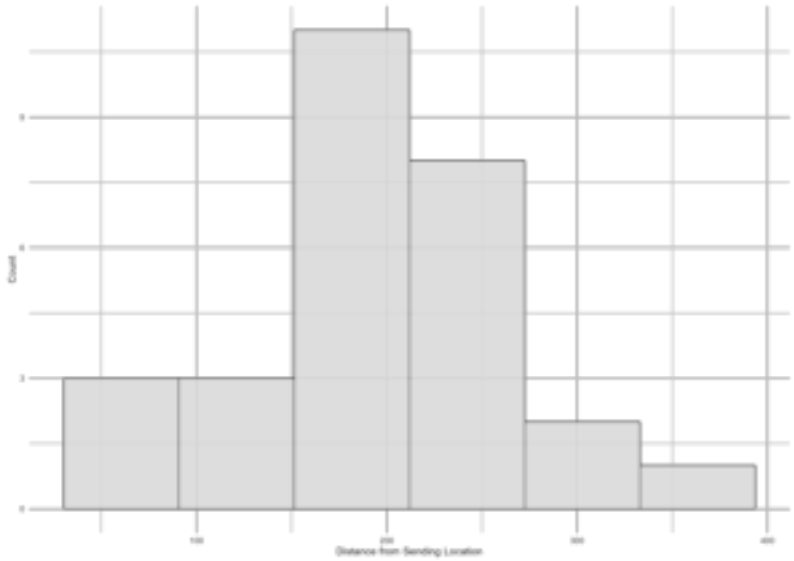


Figure 21. Distance between sending and receiving municipalities. Source: own calculations using road travel times in R.

Appendix C: Empirical Methodology

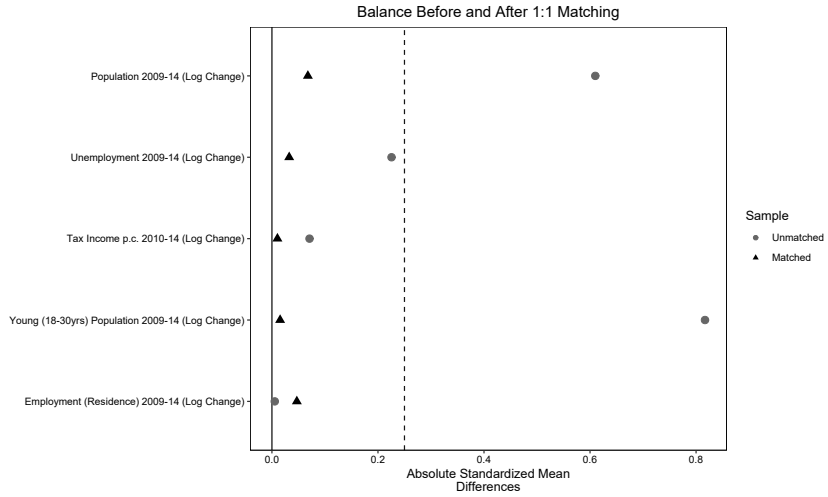


Figure 22. Balance plot showing standardized mean differences and variance ratios between treated municipalities and matched control municipalities inside Bavaria before and after Mahalanobis nearest-neighbor matching.

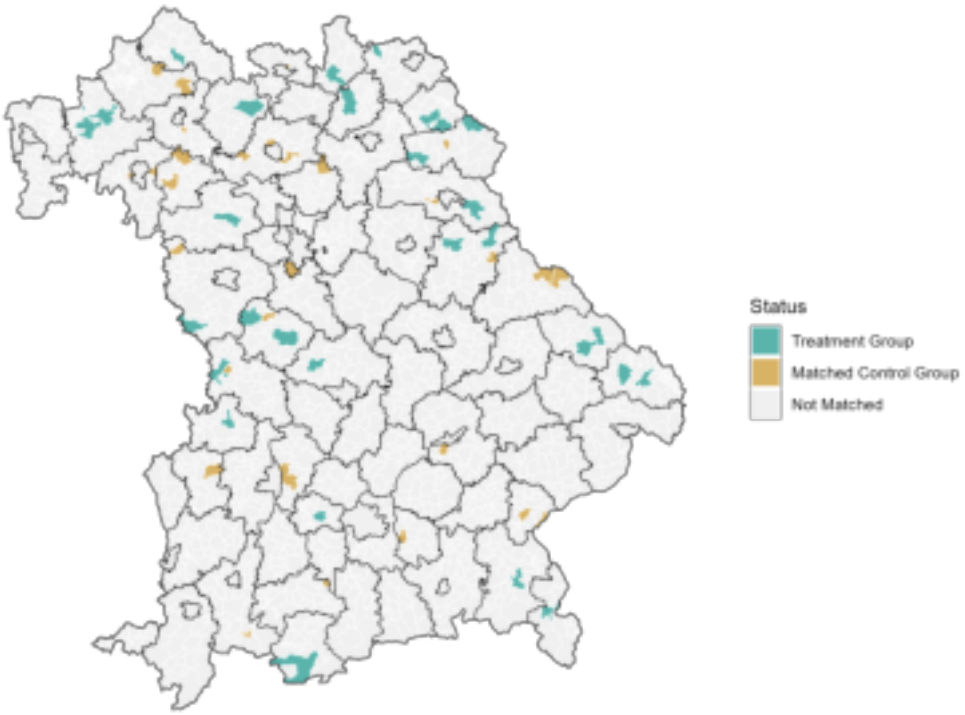


Figure 23. Map of Bavaria showing treated municipalities in teal and their matched control counterparts in Bavaria in ocher.

Variable	Mean Treated	Mean Control	Std. Mean Diff.	Var. Ratio
Population	-0.01	0.00	-0.43	0.67
Unemployment	-0.25	-0.26	0.01	0.35
Employment	0.10	0.09	0.15	0.37
Income per taxpayer	0.14	0.15	-0.22	0.44
Population 18–30 yrs	-0.03	0.01	-0.57	0.37

Table 19. Balance before matching between treated municipalities in Bavaria and other municipalities in Bavaria.

Variable	Mean Treated	Mean Control	Std. Mean Diff.	Var. Ratio
Population	-0.01	-0.01	0.00	1.08
Unemployment	-0.25	-0.25	-0.01	1.08
Employment	0.10	0.09	0.12	1.23
Income per taxpayer	0.14	0.15	-0.05	1.23
Population 18–30 yrs	-0.03	-0.04	0.08	0.85

Table 20. Balance after matching between treated municipalities in Bavaria and other municipalities in Bavaria.

Appendix D: Long-Difference Results with Bavarian Control Group

	2015	2016	2017	2018	2019	2020	2021	2022
Treatment (Bin)	0.0067	0.0080	0.0056	0.0066	0.0032	0.0053	0.0051	0.0055
	(0.0046)	(0.0058)	(0.0072)	(0.0066)	(0.0076)	(0.0091)	(0.0093)	(0.0091)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	76	76	76	76	76	76	76
R^2	0.54864	0.55713	0.44794	0.53607	0.52680	0.51694	0.50185	0.53778
Within R^2	0.08746	0.16938	0.10287	0.12975	0.09933	0.22052	0.15279	0.17164
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in employment shares for endpoint years from 2015 to 2022 relative to 2014. The control group consists of municipalities inside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 21. Long-difference results for employment shares using the Bavarian control group, 2015–2022.

	2015	2016	2017	2018	2019	2020	2021	2022
Treatment (Bin)	-0.0102	-0.0215	-0.0244	-0.0393	-0.0297	0.0438	0.0009	-0.0145
	(0.0331)	(0.0371)	(0.0355)	(0.0360)	(0.0335)	(0.0396)	(0.0351)	(0.0454)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	76	76	76	76	76	76	76
R^2	0.52555	0.58116	0.65309	0.63236	0.60578	0.73557	0.69041	0.68891
Within R^2	0.06125	0.11097	0.25980	0.21547	0.12309	0.27213	0.29037	0.19130
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in unemployment shares for endpoint years from 2015 to 2022 relative to 2014. The control group consists of municipalities inside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 22. Long-difference results for unemployment shares using the Bavarian control group, 2015–2022.

	2015	2016	2017	2018	2019	2020	2021	2022	2023
Treatment (Bin)	0.0027	0.0075*	0.0105	0.0111	0.0149	0.0125	0.0133	0.0224	0.0310*
	(0.0031)	(0.0041)	(0.0064)	(0.0083)	(0.0091)	(0.0101)	(0.0112)	(0.0135)	(0.0168)
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	76	76	76	76	76	76	76	76
R^2	0.63082	0.69660	0.68915	0.69285	0.74585	0.73832	0.71614	0.68532	0.64910
Within R^2	0.07998	0.19086	0.19880	0.19749	0.20461	0.17238	0.17179	0.20876	0.21334
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in working-age population for endpoint years from 2015 to 2023 relative to 2014. The control group consists of municipalities inside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 23. Long-difference results for working-age population using the Bavarian control group, 2015–2023.

	2015	2016	2017	2018	2019	2020	2021	2022
Treatment (Bin)	0.0033	0.0076**	0.0103*	0.0097	0.0095	0.0050	0.0003	0.0056
	(0.0026)	(0.0034)	(0.0054)	(0.0069)	(0.0078)	(0.0083)	(0.0080)	(0.0095)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	76	76	76	76	76	76	76	76
R^2	0.64695	0.66852	0.66129	0.67484	0.70966	0.69278	0.69274	0.63442
Within R^2	0.07340	0.21783	0.23168	0.24875	0.22126	0.16491	0.18066	0.15370
Mahalanobis Pair Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓

Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Notes: The table reports log differences in total population for endpoint years from 2015 to 2022 relative to 2014. The control group consists of municipalities inside Bavaria matched to treated Bavarian municipalities using Mahalanobis distance. Each treated municipality is compared to its Mahalanobis nearest neighbor.

Table 24. Long-difference results for total population using the Bavarian control group, 2015–2022.

Appendix E: Event-Study Results with Bavarian Control Group

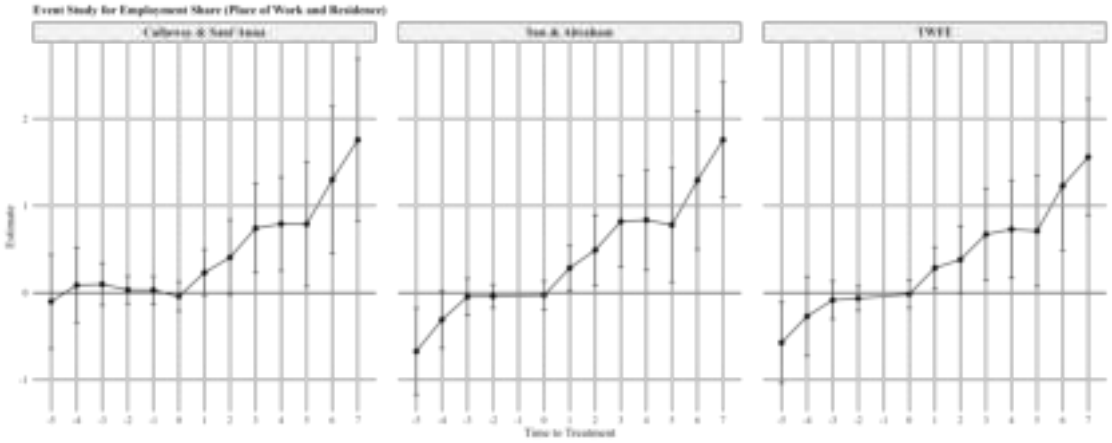


Figure 24. Event-study estimates for employment shares using the Bavarian control group. Employment is measured for individuals who live and work in the same municipality. The figure reports Callaway and Sant'Anna (2021), Sun and Abraham (2021), and TWFE estimates.

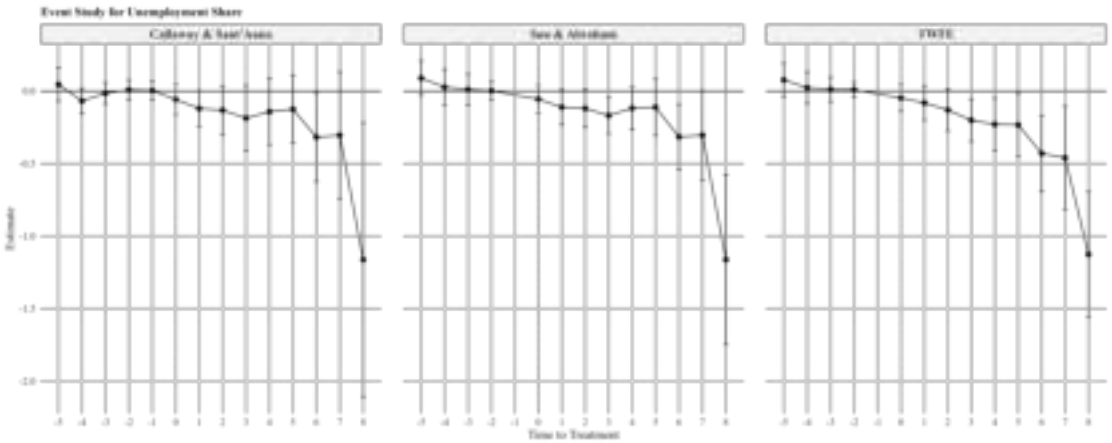


Figure 25. Event-study estimates for unemployment shares using the Bavarian control group. The figure reports Callaway and Sant'Anna (2021), Sun and Abraham (2021), and TWFE estimates.

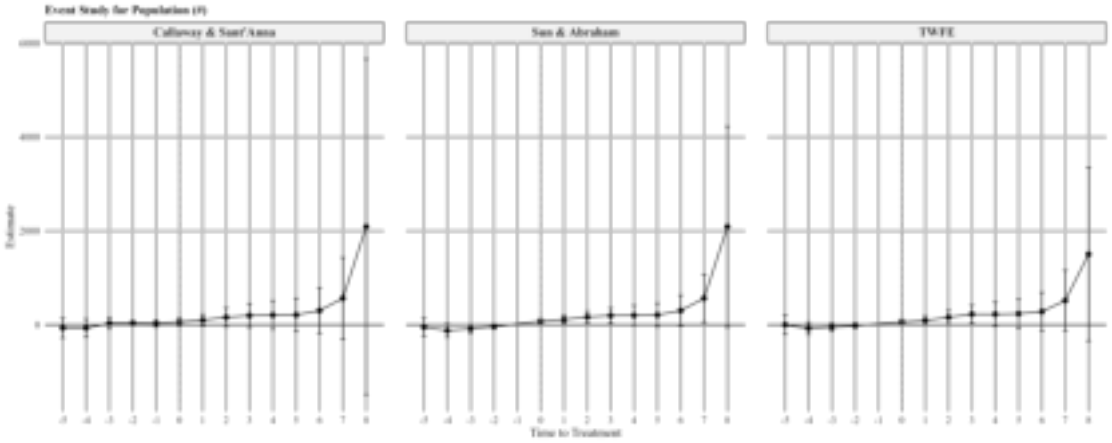


Figure 26. Event-study estimates for total population using the Bavarian control group. The figure reports Callaway and Sant'Anna (2021), Sun and Abraham (2021), and TWFE estimates.

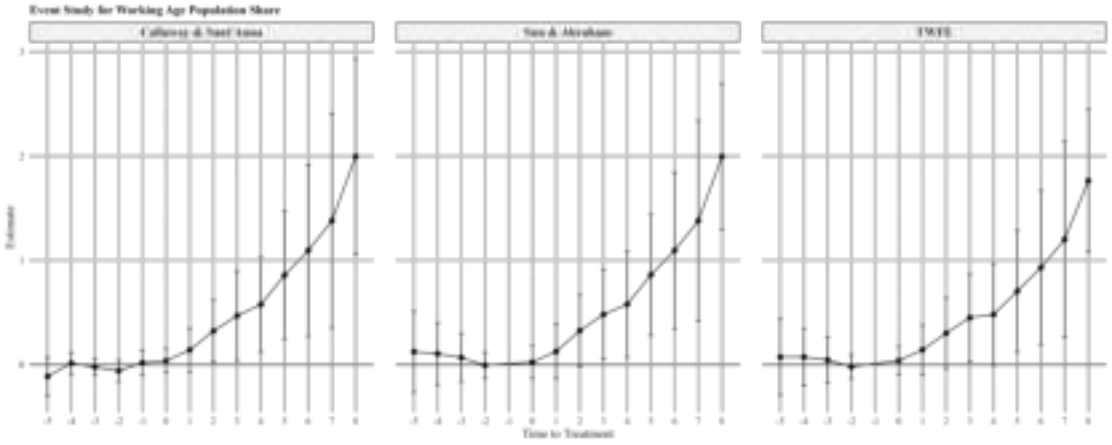


Figure 27. Event-study estimates for the working-age population share using the Bavarian control group. The figure reports Callaway and Sant'Anna (2021), Sun and Abraham (2021), and TWFE estimates.

Appendix F: Recode of the ARSW Toolkit into R

The quantitative spatial model of Ahlfeldt, Redding, et al. (2015) was originally implemented in MATLAB. I recoded the toolkit in R, retaining the modular structure of the original code while adapting it to tidyverse-style workflows and extending it to a two-sector setting with private and state public employment. This appendix documents the recode and provides a crosswalk between the original ARSW implementation and the current R implementation.

Data Preparation. The input data are the 2011 German Census commuting tables, `2000X_2041` for in-commuters and `2000X_2045` for out-commuters, supplemented with geospatial data and socio-economic variables. The scripts parse the Census files, filter the relevant categories, and construct employment variables by workplace and place of residence. For the model calibration, I classify employment in *öffentliche Verwaltung, Verteidigung, Sozialversicherung* as public employment and all other sectors as private employment. The current implementation is at the county level, consistent with the model results reported in the main text and presentation.

Travel Times. Bilateral travel times between locations are computed from OpenStreetMap using the `osrm` package in R. The procedure returns road travel times τ_{ij}, which enter commuting costs as

$$d_{ij} = \exp(\kappa \tau_{ij}).$$

The travel-time matrix is also used to construct distance-weighted productivity and amenity spillovers.

Commuting Parameters. The commuting-cost parameter is estimated using bilateral commuting flows from the Federal Employment Agency. I estimate the gravity equation

$$\log(\text{commuters}_{ij}) = \alpha_i + \gamma_j - (\kappa\epsilon)\tau_{ij} + \varepsilon_{ij},$$

with origin and destination fixed effects. I then recover κ by combining the estimate of $\kappa\epsilon$ with the calibrated Fréchet shape parameter ϵ. In the current county-level implementation, the estimated parameters are

$$\widehat{\epsilon} = 3.11, \qquad \widehat{\kappa} = 0.0034.$$

Recovery of Fundamentals. The recode follows the ARSW inversion logic to recover adjusted productivity and amenities from observed wages, rents, commuting flows, employment, and land area. In the two-sector extension, private-sector productivity is modeled as

$$A_j^P = a_j \left(\Upsilon_j^P\right)^{\lambda} \left(\Upsilon_j^G\right)^{\sigma\lambda},$$

where

$$\Upsilon_j^m = \sum_s \exp(-\delta \tau_{js}) \frac{H_{m,s}^M}{K_s}, \qquad m \in \{P, G\}.$$

Here λ governs within-private-sector spillovers, σ scales public-to-private spillovers relative to within-private-sector spillovers, and δ controls spatial decay. This specification is the one used in the current paper and presentation.

Residential amenities are given by

$$B_i = b_i \Omega_i^\eta, \qquad \Omega_i = \sum_s \exp(-\rho \tau_{is}) \frac{H_s^R}{K_s}.$$

Amenities depend on total resident density and are not sector-specific.

Counterfactuals with Public Employment Reallocation. The counterfactual implements the *Heimatstrategie* as an exogenous reallocation of public workplace employment:

$$\bar{H}_{Mj}^{G,\text{cf}} = \bar{H}_{Mj}^{G,0} + \Delta H_{Mj}^G, \qquad \sum_j \Delta H_{Mj}^G = 0.$$

Public jobs are assigned to workplaces by policy, while public workers choose where to live and commute. The model is re-solved for wages, rents, commuting flows, private workplace employment, residence employment, productivity, and amenities. The current model results use the newest county-level simulations reported in the presentation, with output files stored in `qsm/2026_05_09/`.

Visualization. The plotting module uses `ggplot2` and `sf` to map counterfactual changes in private employment, productivity, amenities, wages, residence employment, total workplace employment, and residential rents. The current paper uses the following simulation outputs:

`single_HP_pct_change_by_plus_county.pdf`,

`single_A_change_by_plus_county.pdf`,

`single_B_change_by_plus_county.pdf`,

`single_w_change_by_plus_county.pdf`,

`single_HM_change_by_plus_county.pdf`,

`single_HR_pct_change_by_plus_county.pdf`,

and

`single_Q_res_change_by_plus_county.pdf`.

Current Status and Next Steps. The model is currently implemented at the county level and uses the 2011 Census as a preliminary source for sectoral workplace and residence employment. The next step is to move the model to the municipal level by completing the yearly municipal employment panel that combines IAB social-security records with civil-servant data by workplace and residence. I will also incorporate commercial floor-space and commercial price data to model the real-estate margin of the policy more explicitly, including the release of office floor space in urban sending locations. Finally, I will estimate the spillover elasticities via GMM, targeting the empirical policy responses as moment conditions.